Lecture 16

Gibbs sampling

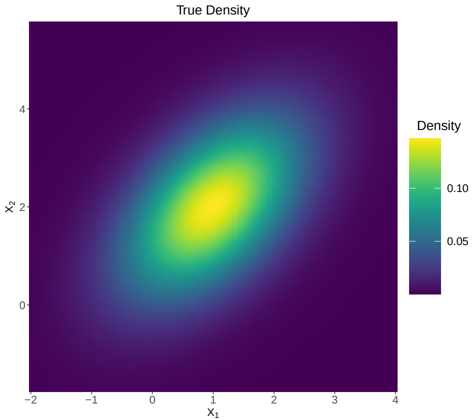
Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Monte Carlo sampling

- Monte Carlo (MC) sampling is the predominant method of Bayesian inference.
- The main reason is that it generally preforms better than other alternatives for high-dimensional models (i.e., with many parameters).
- The main idea is to approximate posterior summaries in terms of some randomly drawn samples from the posterior distribution.
- This often requires drawing samples from non-standard distributions.
- It also requires careful analysis to be sure the approximation is sufficiently accurate.

Example


True Density
Density
0.10
0.05
4
2
0
−2
−1
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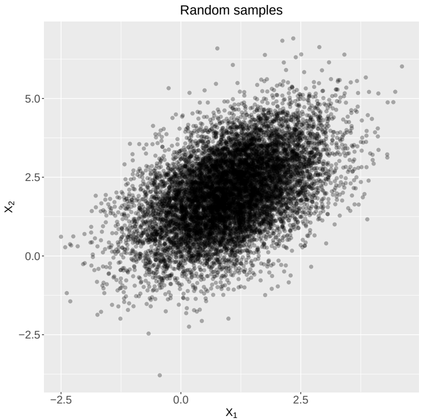
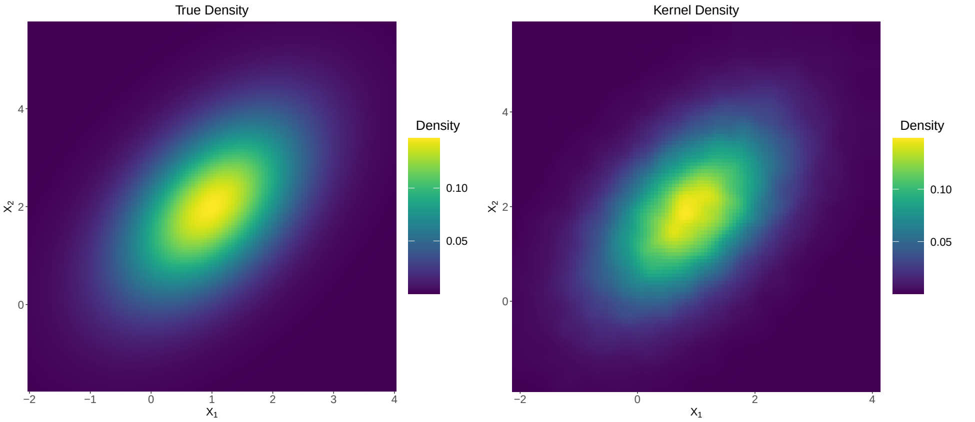
1
2
3
4
X_1
X_2


Random samples
5.0
2.5
0.0
−2.5
−2.5
0.0
2.5
X_1
X_2

Example



True Density
Density
0.10
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3
4
X_1
Kernel Density
Density
0.10
0.05
4
2
0
X_2
−2
0
2
4
X_1

Monte Carlo sampling

- Notation: Let $\boldsymbol{\theta} = (\theta_1, ..., \theta_p)'$ be the collection of all parameters in the model.
- Notation: Let $\mathbf{Y} = (Y_1, ..., Y_n)'$ be entire dataset.
- The posterior $f(\boldsymbol{\theta}|\mathbf{Y})$ is a distribution.
- If $\boldsymbol{\theta}^{(1)}, ..., \boldsymbol{\theta}^{(S)}$ are samples from $f(\boldsymbol{\theta}|\mathbf{Y})$, then the mean of the S samples approximates the posterior mean.
- This only provides approximations of the posterior summaries of interest.
- But how to draw samples from some arbitrary distribution $p(\boldsymbol{\theta}|\mathbf{Y})$?

Software optioms

- There are now many software options for performing MC sampling.
- There are `R` functions for particular analyses (e.g., the function `BLR` for linear regression).
- There are also all-purpose programs that work for virtually any user-specified model: OpenBUGS; JAGS; Proc MCMC; STAN; INLA (not MC).
- We will use JAGS and also STAN (if possible), but they are all similar.

Markov chain Monte Carlo (MCMC)

We will study the algorithms behind these programs, which is important because it helps to

- Select models and priors,
- Anticipate bottlenecks,
- Understand error messages and output,
- Design your own sampler if these off-the-shelf programs are too slow.

The most common algorithms are **Gibbs** and **Metropolis** sampling.

Gibbs sampling

- Gibbs sampling is attractive because it can sample from high-dimensional posteriors.
- It breaks the problem of sampling from a high-dimensional joint distribution into a series of sampling from low-dimensional conditional distributions.
- Updates can also be done in blocks (groups of parameters).
- Because the low-dimensional updates are done in a loop, samples are not independent.
- The dependence turns out to be a Markov distribution, leading to the name Markov chain Monte Carlo (MCMC).

Gibbs sampling for the Gaussian model

- Say $Y_i \overset{IID}{\sim} \text{Normal}(\mu, \sigma^2)$ with $\mu \sim \text{Normal}(0, \sigma_0^2)$, and $\sigma^2 \sim \text{InvGamma}(a, b)$.
- In Chapter 2 (while discussing conjugate priors), we saw that if we knew either μ or σ^2, we can sample from the other parameter.
- $\mu|\sigma^2, \mathbf{Y} \sim \text{Normal}\left(\frac{n\bar{Y}\sigma^{-2}+\mu_0\sigma_0^{-2}}{n\sigma^{-2}+\sigma_0^{-2}}, \frac{1}{n\sigma^{-2}+\sigma_0^{-2}}\right)$.
- $\sigma^2|\mu, \mathbf{Y} \sim \text{Inverse-Gamma}\left(\frac{n}{2} + a, \frac{1}{2}\sum_{i=1}^{n}(Y_i - \mu)^2 + b\right)$.
- But how to draw from the joint distribution $\pi(\mu, \sigma^2|\mathbf{Y})$?

Gibbs sampling for the Gaussian model

- The full conditional (FC) distribution is the distribution of one parameter taking all other as fixed and known.
- FC1: $\mu|\sigma^2, \mathbf{Y} \sim \text{Normal}\left(\frac{n\bar{Y}\sigma^{-2}+\mu_0\sigma_0^{-2}}{n\sigma^{-2}+\sigma_0^{-2}}, \frac{1}{n\sigma^{-2}+\sigma_0^{-2}}\right)$
- FC2: $\sigma^2|\mu, \mathbf{Y} \sim \text{Inverse-Gamma}\left(\frac{n}{2}+a, \frac{1}{2}\sum_{i=1}^{n}(Y_i-\mu)^2+b\right)$

Gibbs sampling

- In the Gaussian model $\boldsymbol{\theta} = (\mu, \sigma^2)$ so $\theta_1 = \mu$ and $\theta_2 = \sigma^2$.
- The algorithm begins by setting initial values for all parameters, $\boldsymbol{\theta}^{(0)} = (\theta_1^{(0)}, ..., \theta_p^{(0)})$.
- Parameters are then sampled one at a time from their full conditional distributions,
$$p(\theta_j | \theta_1, ..., \theta_{j-1}, \theta_{j+1}, ..., \theta_p, \mathbf{Y}).$$
- Rather than 1 p-dimensional joint sample, we make p 1-dimensional samples.
- The process is repeated until the required number of samples have been generated.

Gibbs sampling

A Set initial value $\boldsymbol{\theta}^{(0)} = (\theta_1^{(0)}, ..., \theta_p^{(0)})$

B For iteration t,

FC1 Draw $\theta_1^{(t)} | \theta_2^{(t-1)}, ..., \theta_p^{(t-1)}, \mathbf{Y}$

FC2 Draw $\theta_2^{(t)} | \theta_1^{(t)}, \theta_3^{(t-1)}, ..., \theta_p^{(t-1)}, \mathbf{Y}$

...

FCp Draw $\theta_p^{(t)} | \theta_1^{(t)}, ..., \theta_{p-1}^{(t)}, \mathbf{Y}$

We repeat step B S times and get the posterior draws

$$\boldsymbol{\theta}^{(1)}, ..., \boldsymbol{\theta}^{(S)}.$$

Why does this work?

- $\theta^{(0)}$ isn't a sample from the posterior and it is an arbitrarily chosen initial value.
- $\theta^{(1)}$ isn't likely from the posterior either. Its distribution depends on $\theta^{(0)}$.
- $\theta^{(2)}$ isn't likely from the posterior either. Its distribution depends on $\theta^{(0)}$ and $\theta^{(1)}$.
- **Theorem**: For any initial values, the chain will eventually converge to the posterior.
- **Theorem**: If $\theta^{(s)}$ is a sample from the posterior, then $\theta^{(s+1)}$ is too.

Convergence

- We need to decide:
  1. When has it converged?
  2. When have we taken enough samples to approximate the posterior?
- Once we decide the chain has converged at iteration T, we discard the first T samples as "burn-in".

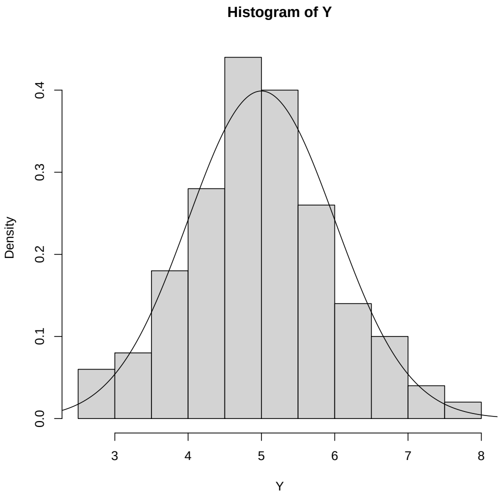
- We use the remaining $S - T$ to approximate the posterior.

- For example, the posterior mean (marginal over all other parameters) of θ_j is

$$\mathrm{E}(\theta_j | \mathbf{Y}) \approx \frac{1}{S - T} \sum_{s=S-T+1}^{S} \theta_j^{(s)}.$$

Let's code up (in R)

```
mu <- 5
sigmaSq <- 1

set.seed(100)
Y <- rnorm(100, mean = mu,
                sd = sqrt(sigmaSq))

hist(Y, probability = T)
grid.vals <- seq(mu - 4, mu + 4, 0.01)
y.vals <- dnorm(grid.vals, mean = mu,
                sd = sqrt(sigmaSq))
lines(grid.vals, y.vals)
```



Let's code up (in R)

- FC1: $\mu | \sigma^2, \mathbf{Y} \sim \text{Normal}\left(\frac{n\bar{Y}\sigma^{-2} + \mu_0\sigma_0^{-2}}{n\sigma^{-2} + \sigma_0^{-2}}, \frac{1}{n\sigma^{-2} + \sigma_0^{-2}}\right)$
- FC2: $\sigma^2 | \mu, \mathbf{Y} \sim \text{Inverse-Gamma}\left(\frac{n}{2} + a, \frac{1}{2}\sum_{i=1}^{n}(Y_i - \mu)^2 + b\right)$

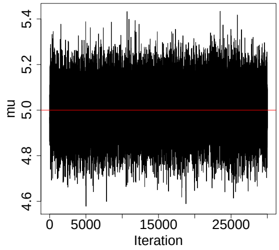
```
mu.update <- function(Y, sigmaSq, mu0, sigmaSq0){
  mu.posvar.inv <- length(Y) / sigmaSq + 1 / sigmaSq0
  mu.posvar <- 1 / mu.posvar.inv
  mu.posmean <- (sum(Y) / sigmaSq + mu0 / sigmaSq0) / mu.posvar.inv
  out <- mu.posmean + sqrt(mu.posvar) * rnorm(1)
  out}

sigmaSq.update <- function(Y, mu, a, b){
  sigmaSq.inv <- rgamma(1, shape = a + length(Y) / 2,
                        rate = b + sum((Y - mu)^2) / 2)
  sigmaSq <- 1 / sigmaSq.inv
  sigmaSq}
```

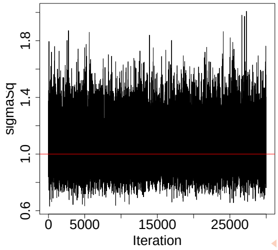
Let's code up (in R)

```
MCMC <- function(Y, mu.init, sigmaSq.init, mu0, sigmaSq0, a, b, iters){

  # chain initiation
  mu <- mu.init
  sigmaSq <- sigmaSq.init

  # define chains
  mu.chain <- rep(NA, iters)
  sigmaSq.chain <- rep(NA, iters)

  # start MCMC
  for(i in 1:iters){
    mu <- mu.update(Y, sigmaSq, mu0, sigmaSq0)
    sigmaSq <- sigmaSq.update(Y, mu, a, b)

    mu.chain[i] <- mu
    sigmaSq.chain[i] <- sigmaSq
  }

  # return chains
  out <- list(mu.chain = mu.chain,
              sigmaSq.chain = sigmaSq.chain)
  return(out)}
```

Let's code up (in R)

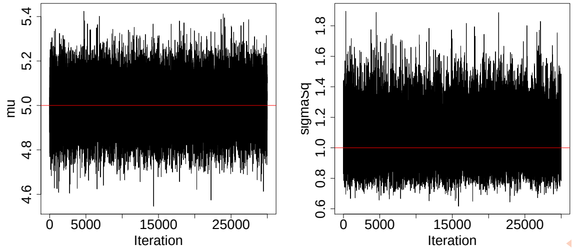
```
MCMC.out <- MCMC(Y = Y,
                 mu.init = mean(Y),
                 sigmaSq.init = var(Y),
                 mu0 = 0,
                 sigmaSq0 = 100,
                 a = 0.1,
                 b = 0.1,
                 iters = 30000)

plot(MCMC.out$mu.chain, xlab = "Iteration", ylab = "mu", type = "l")
plot(MCMC.out$sigmaSq.chain, xlab = "Iteration", ylab = "sigmaSq", type = "l")
```





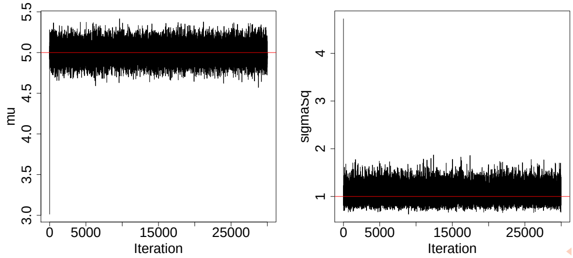
Let's code up (in R)

```r
MCMC.out <- MCMC(Y = Y,
                 mu.init = -10,
                 sigmaSq.init = 10,
                 mu0 = 0,
                 sigmaSq0 = 100,
                 a = 0.1,
                 b = 0.1,
                 iters = 30000)

plot(MCMC.out$mu.chain, xlab = "Iteration", ylab = "mu", type = "l")
plot(MCMC.out$sigmaSq.chain, xlab = "Iteration", ylab = "sigmaSq", type = "l")
```



Let's code up (in R)

```
MCMC.out <- MCMC(Y = Y,
                 mu.init = -1000,
                 sigmaSq.init = 1000,
                 mu0 = 0,
                 sigmaSq0 = 100,
                 a = 0.1,
                 b = 0.1,
                 iters = 30000)

plot(MCMC.out$mu.chain, xlab = "Iteration", ylab = "mu", type = "l")
plot(MCMC.out$sigmaSq.chain, xlab = "Iteration", ylab = "sigmaSq", type = "l")
```



Another example

$Y|\lambda, b \sim \text{Poisson}(\lambda)$, $\lambda|b \sim \text{Exponential}(b)$, $b \sim \text{Exponential}(1)$.

- The full conditional for λ is

$$\begin{aligned} p(\lambda|b, Y) &\propto \frac{f(Y, \lambda, b)}{f(Y, b)} \propto f(Y, \lambda, b) \\ &\propto f(Y|\lambda, b)\pi(\lambda|b)\pi(b) \\ &\propto f(Y|\lambda)\pi(\lambda|b) \\ &\propto \exp[-\lambda]\lambda^{Y} \times b\exp[-b\lambda] \\ &\propto \lambda^{Y+1-1}\exp[-(1+b)\lambda]. \end{aligned}$$

- Therefore, $\lambda|b, Y \sim \text{Gamma}(Y+1, b+1)$.
- In the similar way, $b|\lambda, Y \sim \text{Gamma}(2, \lambda+1)$.

Thank you!

Lecture 17

Metropolis-Hastings sampling

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Recap: Monte Carlo sampling

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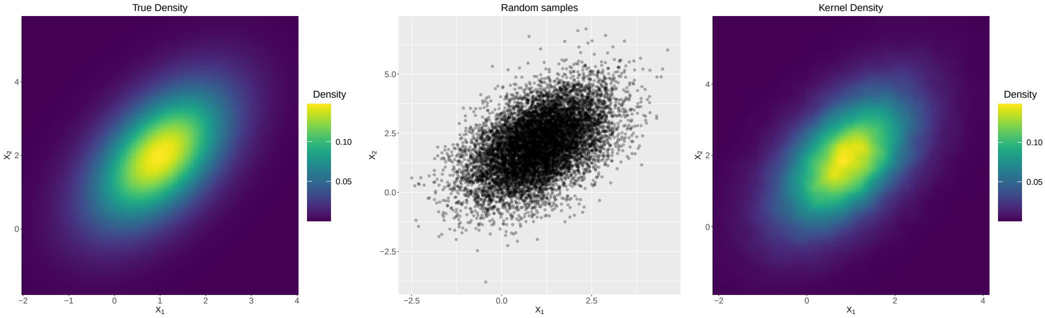
Example



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...

FCp Draw $\theta_p^{(t)} | \theta_1^{(t)}, ..., \theta_{p-1}^{(t)}, \mathbf{Y}$

We repeat step B S times and get the posterior draws

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Recap: Gibbs sampling for the Gaussian model

- Say $Y_i \overset{IID}{\sim} \text{Normal}(\mu, \sigma^2)$ with $\mu \sim \text{Normal}(0, \sigma_0^2)$, and $\sigma^2 \sim \text{InvGamma}(a, b)$.
- In Chapter 2 (while discussing conjugate priors), we saw that if we knew either μ or σ^2, we can sample from the other parameter.
- $\mu | \sigma^2, \mathbf{Y} \sim \text{Normal}\left(\frac{n\bar{Y}\sigma^{-2} + \mu_0\sigma_0^{-2}}{n\sigma^{-2} + \sigma_0^{-2}}, \frac{1}{n\sigma^{-2} + \sigma_0^{-2}}\right)$.
- $\sigma^2 | \mu, \mathbf{Y} \sim \text{Inverse-Gamma}\left(\frac{n}{2} + a, \frac{1}{2}\sum_{i=1}^{n}(Y_i - \mu)^2 + b\right)$.
- But how to draw from the joint distribution $\pi(\mu, \sigma^2 | \mathbf{Y})$? Using Gibbs sampling.

Metropolis sampling

- In Gibbs sampling, each parameter is updated by sampling from its full conditional distribution.
- This is only possible with conjugate priors.
- However, if the prior is not conjugate, it is not obvious how to make a draw from the full conditional.
- For example, if $Y \sim \text{Normal}(\mu, 1)$ and $\mu \sim \text{Beta}(a, b)$ then

$$p(\mu|Y) \propto \exp\left[-\frac{1}{2}(Y-\mu)^2\right]\mu^{(a-1)}(1-\mu)^{b-1}.$$

- For some likelihoods, there is no known conjugate prior, e.g., logistic regression.
- In these cases, we use Metropolis sampling.

Metropolis sampling

- Metropolis sampling is a version of rejection sampling.
- Let θ_j^* be the current value of the parameter being updated and $\theta_{(j)}$ be the current value of all other parameters.
- You propose a random candidate based on the current value, e.g.,

$$\theta_j^c \sim \text{Normal}(\theta_j^*, s_j^2).$$

- The candidate is accepted with probability

$$R = \min\left\{1, \frac{p(\theta_j^c|\theta_{(j)}, \mathbf{Y})}{p(\theta_j^*|\theta_{(j)}, \mathbf{Y})}\right\}.$$

- If the candidate is not accepted then you simply retain the previous value and move to the next step.

Metropolis sampling

- The candidate standard deviation s_j is a tuning parameter.
- Ideally s_j is tuned to give acceptance probability around 0.3–0.4.
- If s_j is too small: Acceptance rate is large but the algorithm fails to explore the posterior well.
- If s_j is too large: Acceptance rate is poor while the algorithm again fails to explore the main important region of the posterior well.
- Off-the-shelf programs have default values, and many allow you to change the value if the results are unsatisfactory.

Metropolis-Hastings sampling

- Denote $\theta_j^c \sim q(\theta|\theta^*)$ as the candidate distribution.
- The candidate distribution is symmetric if

$$q(\theta^*|\theta_j^c) = q(\theta_j^c|\theta^*).$$

- For example, if $\theta_j^c \sim \text{Normal}(\theta_j^*, s_j^2)$ then

$$q(\theta_j^c|\theta^*) = \frac{1}{\sqrt{2\pi}s_j} \exp\left[-\frac{(\theta_j^c - \theta_j^*)^2}{2s_j^2}\right] = q(\theta^*|\theta_j^c).$$

Metropolis-Hastings sampling

- Metropolis-Hastings (MH) sampling generalizes Metropolis sampling to allow for asymmetric candidate distributions.

- For example, if $\theta_j \in [0,1]$ then a reasonable candidate is

$$\theta_j^c | \theta_j^* \sim \text{Beta}[10\theta_j^*, 10(1-\theta_j^*)].$$

- Then $q(\theta_j^* | \theta_j^c)$ and $q(\theta_j^c | \theta^*)$ are both beta PDFs.

- MH proceeds exactly like Metropolis except the acceptance probability is

$$R = \min\left\{1, \frac{p(\theta_j^c | \theta_{(j)}, \mathbf{Y}) q(\theta_j^* | \theta_j^c)}{p(\theta_j^* | \theta_{(j)}, \mathbf{Y}) q(\theta_j^c | \theta_j^*)}\right\}.$$

Metropolis-Hastings sampling

- What if we take the candidate distribution to be the full conditional distribution

$$\theta_j^c \sim p(\theta_j^c|\theta_{(j)}, \mathbf{Y})?$$

- What is the acceptance ratio?

$$\frac{p(\theta_j^c|\theta_{(j)}, \mathbf{Y})q(\theta_j^*|\theta_j^c)}{p(\theta_j^*|\theta_{(j)}, \mathbf{Y})q(\theta_j^c|\theta_j^*)} = \frac{p(\theta_j^c|\theta_{(j)}, \mathbf{Y})p(\theta_j^*|\theta_{(j)}, \mathbf{Y})}{p(\theta_j^*|\theta_{(j)}, \mathbf{Y})p(\theta_j^c|\theta_{(j)}, \mathbf{Y})} = 1.$$

- What does this say about the relationship between Gibbs and Metropolis Hastings sampling?

- Gibbs sampling is a special case of the Metropolis-Hastings sampling where the candidate distribution is same as the full conditional distribution.

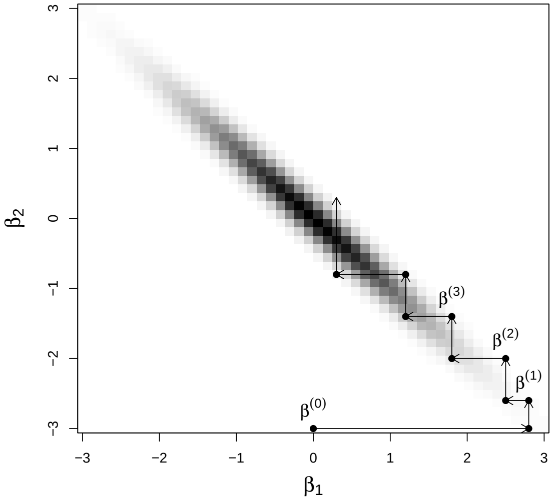
Variants

- You can combine Gibbs and Metropolis in the obvious way, sampling directly from full conditional when possible and Metropolis otherwise.
- Adaptive MCMC varies the candidate distribution throughout the chain.
- Hamiltonian MCMC uses the gradient of the posterior in the candidate distribution and is used in STAN.

Blocked Gibbs/Metropolis

- If a group of parameters are highly correlated, convergence can be slow.
- One way to improve Gibbs sampling is a block update.
- For example, in linear regression might iterate between sampling the block $(\beta_1, \ldots, \beta_p)$ and σ^2.
- Blocked Metropolis is possible too.
- For example, the candidate for $(\beta_1, \ldots, \beta_p)$ could be a multivariate normal.

Posterior correlation leads to slow convergence



Summary

- With the combination of Gibbs and Metropolis-Hastings sampling, we can fit virtually any model.
- In some cases, Bayesian computing is actually preferable to maximum likelihood analysis.
- In most cases Bayesian computing is slower.
- However, in the opinion of many it is worth the wait for improved uncertainty quantification and interpretability.
- In all cases, it is important to carefully monitor convergence.

Thank you!

Lecture 18

MCMC Convergence Diagnostics

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Recap: Gibbs sampling

A Set initial value $\boldsymbol{\theta}^{(0)} = (\theta_1^{(0)}, ..., \theta_p^{(0)})$

B For iteration t,

FC1 Draw $\theta_1^{(t)} | \{\theta_2^{(t-1)}, ..., \theta_p^{(t-1)}, \mathbf{Y}\}$ from the full conditional posterior $f(\theta_1 | rest)$

FC2 Draw $\theta_2^{(t)} | \{\theta_1^{(t)}, \theta_3^{(t-1)}, ..., \theta_p^{(t-1)}, \mathbf{Y}\}$ from the full conditional posterior $f(\theta_2 | rest)$

...

FCp Draw $\theta_p^{(t)} | \{\theta_1^{(t)}, ..., \theta_{p-1}^{(t)}, \mathbf{Y}\}$ from the full conditional posterior $f(\theta_p | rest)$

We repeat step B S times and get the posterior draws

$$\boldsymbol{\theta}^{(1)}, ..., \boldsymbol{\theta}^{(S)}.$$

Recap: Metropolis-Hastings sampling

- MH sampling is a version of rejection sampling.
- Let θ_j^* be the current value of the parameter being updated and $\theta_{(j)}$ be the current value of all other parameters.
- You propose a random candidate based on the current value, $\theta_j^c \sim q(\theta_j|\theta_j^*)$.
- The candidate is accepted with probability

$$R = \min\left\{1, \frac{p(\theta_j^c|\theta_{(j)}, \mathbf{Y})q(\theta_j^*|\theta_j^c)}{p(\theta_j^*|\theta_{(j)}, \mathbf{Y})q(\theta_j^c|\theta_j^*)}\right\}.$$

- If the candidate is not accepted then you simply retain the previous value and move to the next step.

Recap: Why does MCMC work?

- $\theta^{(0)}$ is not a sample from the posterior and it is an arbitrarily chosen initial value.
- $\theta^{(1)}$ is also unlikely from the posterior. Its distribution depends on $\theta^{(0)}$.
- $\theta^{(2)}$ is also unlikely from the posterior. Its distribution depends on $\theta^{(0)}$ and $\theta^{(1)}$.
- **Theorem**: For any initial values, the chain will eventually converge to the posterior.
- **Theorem**: If $\theta^{(s)}$ is a sample from the posterior, then $\theta^{(s+1)}$ is too.

Tuning the MCMC algoritm

- MCMC is beautiful because it can handle virtually any statistical model and it is usually pretty easy to code.
- However, for hard problems great care must be taken to ensure that the algorithm has converged.
- There are three main decisions:
  - Selecting the initial values
  - Determining if/when the chain(s) has converged
  - Selecting the number of samples needed to approximate the posterior

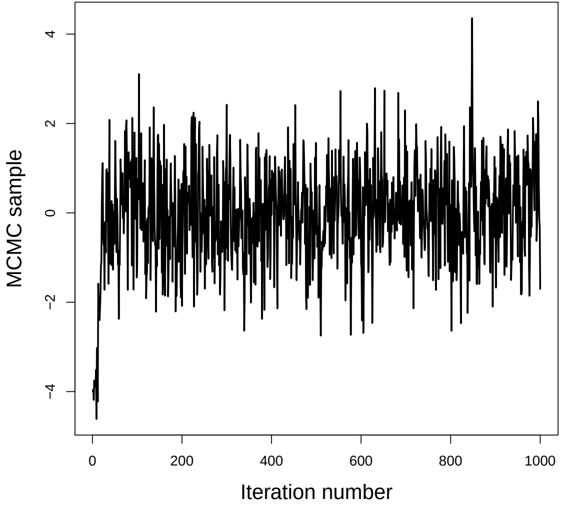
Initial values

- The algorithm will eventually converge no matter what initial values you select.
- However taking time to select good initial values will speed up convergence.
- It is important to try a few initial values to verify they all give the same result.
- Usually 3-5 separate chains is sufficient.
- **Option 1**: Select good initial values using method of moments or MLE.
- **Option 2**: Purposely pick bad but different initial values for each chain to check convergence.

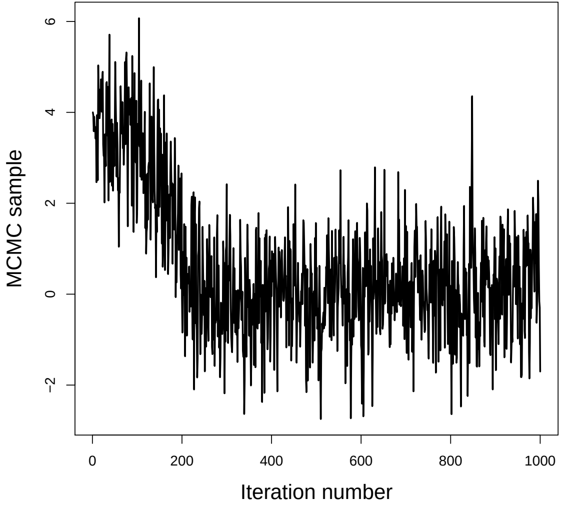
Convergence

- It can take hundreds or even thousands of iterations to move from the initial values to the posterior.
- When the sampler reaches the posterior this is called convergence.
- Samples before convergence are discard as **burn-in**.
- After convergence, the samples should not converge to a single point!
- They should be draws from the posterior, and ideally look like a caterpillar.

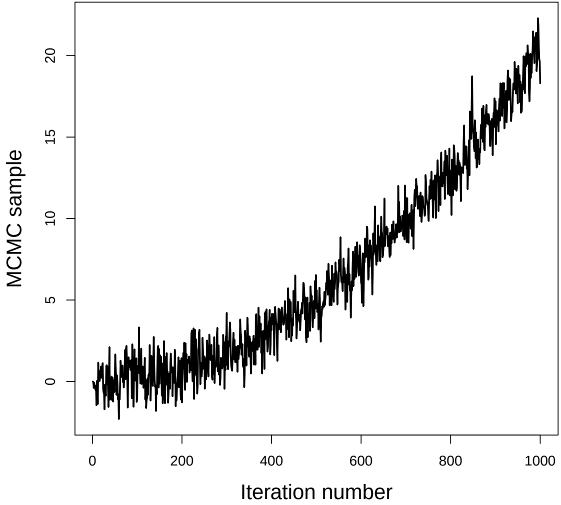
Convergence in a few iterations



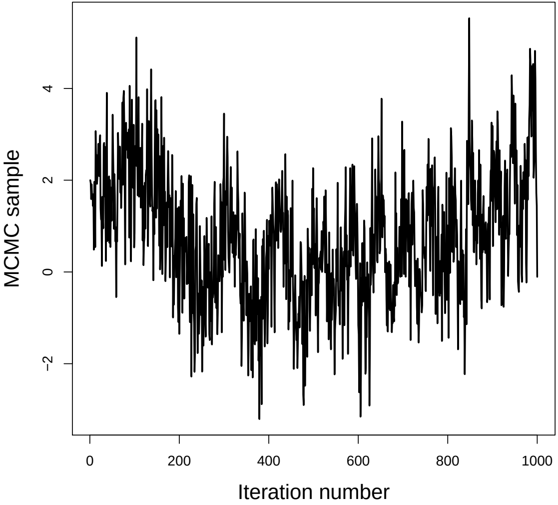
Convergence in a few hundred iterations



This one never converged



Convergence is questionable



Convergence diagnostics

- So far we have visually inspected the chains for convergence.
- There are many formal diagnostics.
- The `CODA` package in `R` has dozens of diagnostics.
- Most give a measure of convergence for each parameter.
- Checking convergence using these one-number summaries is more efficient and objective than visual inspection.

Convergence diagnostics

- Did my chains converge?
  - Geweke
  - Gelman-Rubin
- Did I run the sampler long enough after convergence?
  - Effective sample size
  - Standard errors for the posterior mean estimate

Examples

- The JAGS function `coda.samples` returns sample is the format that can be passed to the `CODA` function which actually computes the diagnostics.

- We will use `CODA` to access convergence for a best-case and a worst-case scenario.

Geweke diagnostic

- It compares the mean in the beginning of the chain (after burn-in period) with the mean at the end of the chain.
- This can be used for a single chain.
- This is done separately for each parameter.
- The JAGS default is to compare the first 10% with the last 50%.
- The test accounts for the inherent autocorrelation in MCMC.
- The test statistic is a z-score, so $|Z| > 2$ indicates poor convergence.

Gelman-Rubin statistic

- If we run multiple chains, we hope that all chains give same result.
- The Gelman-Rubin statistics measures agreement between chains.
- Is it essentially an ANOVA test of whether the chains have the same mean.
- It is scaled so that 1 is perfect and 1.1 is decent but not great convergence.
- JAGS plots the statistic over iteration.
- When the statistic reaches one, this indicates convergence.

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Revisiting the Gelman–Rubin Diagnostic

Dootika Vats and Christina Knudson

Abstract. Gelman and Rubin's (*Statist. Sci.* **7** (1992) 457–472) convergence diagnostic is one of the most popular methods for terminating a Markov chain Monte Carlo (MCMC) sampler. Since the seminal paper, researchers have developed sophisticated methods for estimating variance of Monte Carlo averages. We show that these estimators find immediate use in the Gelman–Rubin statistic, a connection not previously established in the literature. We incorporate these estimators to upgrade both the univariate and multivariate Gelman–Rubin statistics, leading to improved stability in MCMC termination time. An immediate advantage is that our new Gelman–Rubin statistic can be calculated for a single chain. In addition, we establish a one-to-one relationship between the Gelman–Rubin statistic and effective sample size. Leveraging this relationship, we develop a principled termination criterion for the Gelman–Rubin statistic. Finally, we demonstrate the utility of our improved diagnostic via examples.

Autocorrelation

- Ideally the samples should be independent across iteration.
- The autocorrelation function $\rho(h)$ is the correlation between samples h iterations apart.
- JAGS plots the autocorrelation as a function of h.
- Lower values are better, but if the chains are long enough even large values can be OK.
- **Thinning**: If autocorrelation is (close to) zero after lag h, you can thin the samples by h to achieve (approximate) independence.
- This is always less efficient than using all samples, but can save memory.

Effective sample size

- Highly correlated samples have less information than independent samples.
- Say S is the actual number of MCMC samples.
- The **effective samples size** is

$$ESS = \frac{S}{1 + 2\sum_{h=1}^{\infty} \rho(h)}.$$

- The correlated MCMC sample of length S has the same information as ESS independent samples.
- In practice, ESS should be at least a few thousand for all parameters.

Standard errors of posterior mean estimates

- The sample mean of the MCMC draws is an estimate of the posterior mean.
- The standard error of this estimate as another diagnostic.
- Assuming independence the standard error is

$$\text{Naive SE} = \frac{s}{\sqrt{S}},$$

where s is the sample SD and S is the number of samples.

- A more realistic standard error is

$$\text{Times-series SE} = \frac{s}{\sqrt{ESS}}.$$

Thank you!

MCMC success example

Arnab Hazra

2/27/2022

We first simulate two sets of observations $X_1, \ldots, X_m \sim \text{Normal}(\mu_X, 1)$ and $Y_1, \ldots, Y_n \sim \text{Normal}(\mu_Y, 1)$. We want to estimate μ_X and μ_Y. We choose priors $\mu_X, \mu_Y \sim \text{Normal}(0, 100^2)$.

```r
rm(list = ls())

m <- 100
n <- 100

mu.X <- 2
mu.Y <- 4

set.seed(100)

X <- rnorm(m, mean = mu.X, sd = 1)
Y <- rnorm(n, mean = mu.Y, sd = 1)

data <- list(X = X, Y = Y, m = m, n = n)
```

Next, we write the script for JAGS.

(1) Define the model as a string.

```r
library(rjags)
```

```
## Warning: package 'rjags' was built under R version 4.0.5
```

```
## Loading required package: coda
```

```
## Linked to JAGS 4.3.0
```

```
## Loaded modules: basemod,bugs
```

```r
model_string <- textConnection("model{

  # Likelihood (dnorm uses a precision, not variance)
  for(i in 1:m){
  X[i] ~ dnorm(mu.X, 1)
  }
  for(j in 1:n){
  Y[j] ~ dnorm(mu.Y, 1)
  }

  # Priors

  mu.X ~ dnorm(0, 0.0001)
```

```
  mu.Y ~ dnorm(0, 0.0001)
  }")
```

(2) Load the data and compile the MCMC code

```
inits <- list(mu.X = mean(X), mu.Y = mean(Y))
model <- jags.model(model_string, data = data, inits = inits, quiet = TRUE, n.chains = 2)
```

(3) Burn-in for 1000 samples

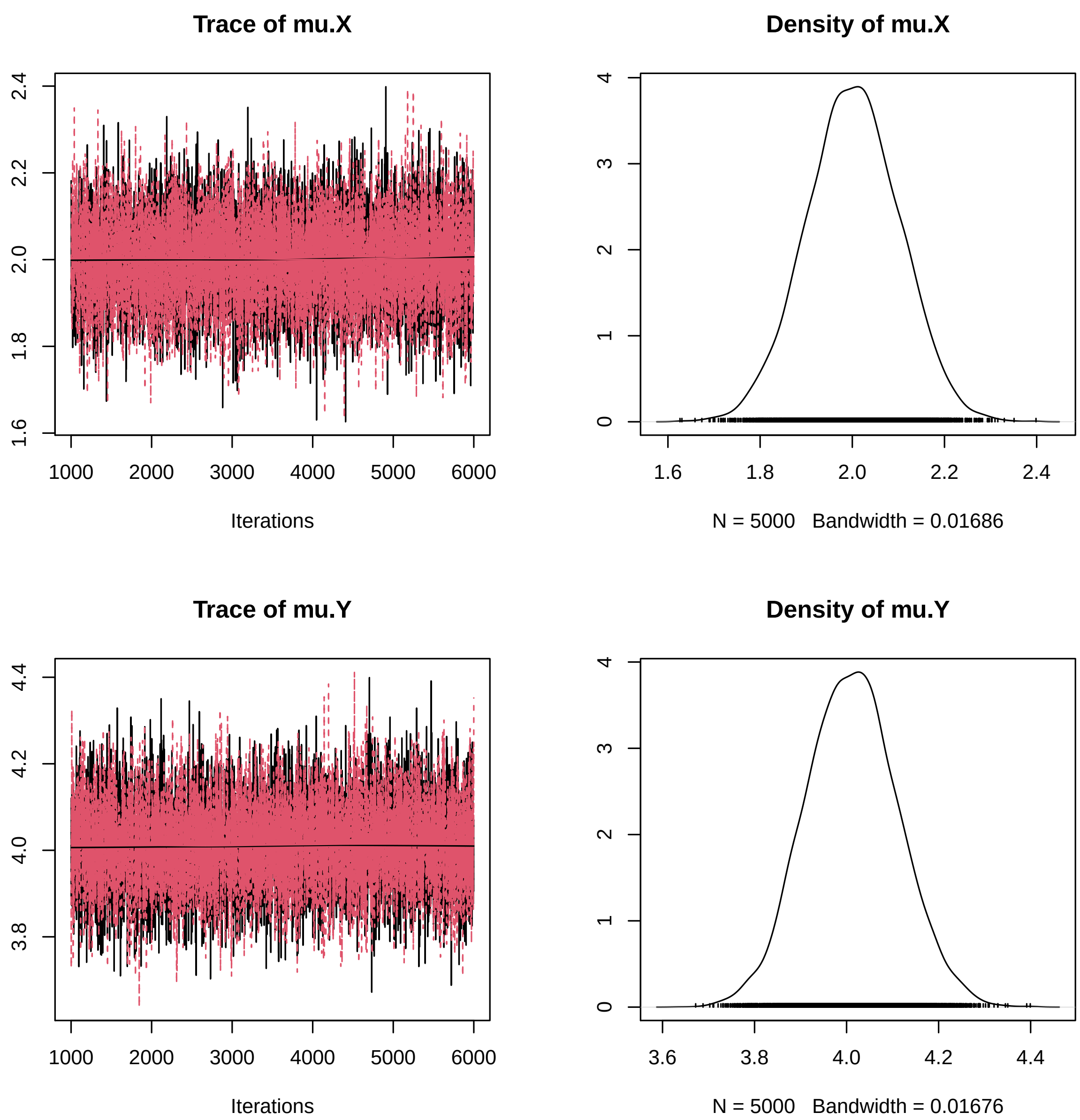
```
update(model, 1000, progress.bar = "none")
```

(4) Generate 5000 post-burn-in samples

```
params <-c("mu.X", "mu.Y")
samples <- coda.samples(model,
                        variable.names = params,
                        n.iter = 5000, progress.bar = "none")
```

(5) Graphical diagnostics

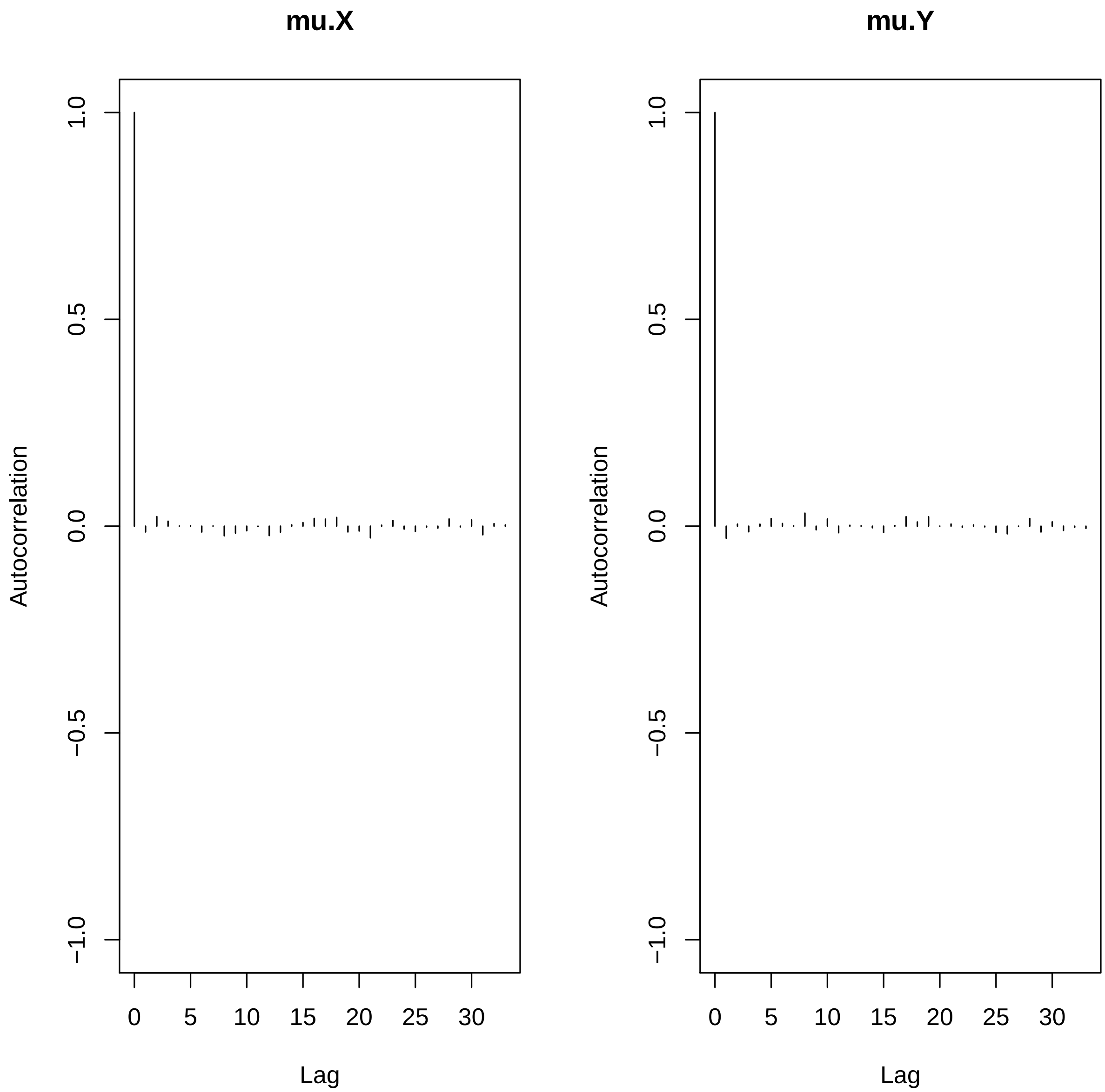
```
plot(samples)
```

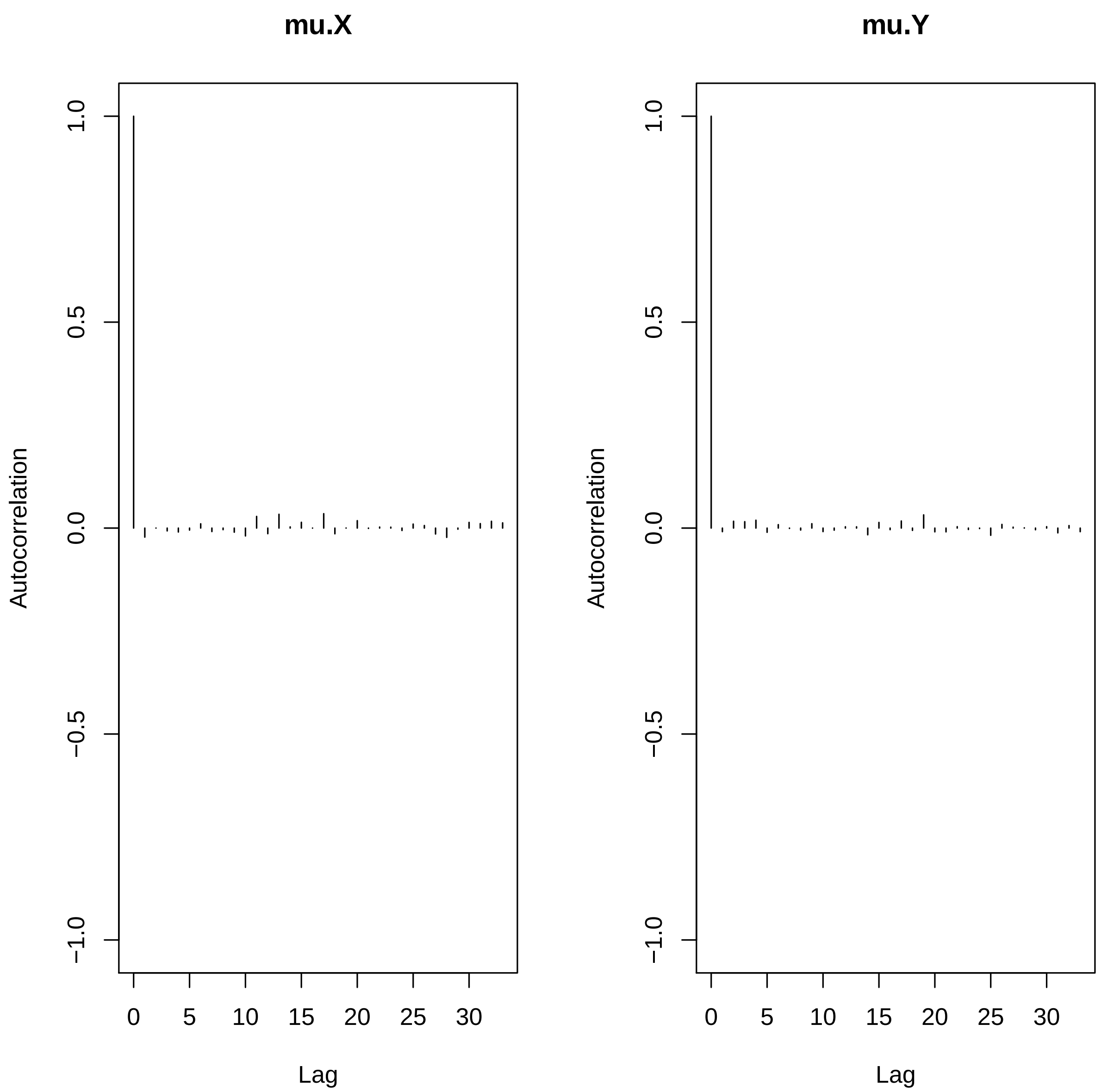


The trace plots look like caterpillars, the chains (one in black and one in gray) give similar estimates.

```
library(coda)

autocorr.plot(samples)
```


mu.X
mu.Y
Autocorrelation
Lag
1.0
0.5
0.0
−0.5
−1.0
0
5
10
15
20
25
30



Autocorrelation is near zero for all lags. Together, these indicate convergence.

Numerical diagnostics

```
# Low autocorrelation indicates convergence

autocorr(samples[[1]], lag = 1)
```

```
## , , mu.X
##
##              mu.X        mu.Y
## Lag 1 -0.0140107 0.005160406
##
## , , mu.Y
##
```

```
##              mu.X        mu.Y
## Lag 1 0.02106957 -0.02915034
```

```r
# high ESS indicates convergence

effectiveSize(samples)
```

```
##     mu.X     mu.Y
## 10223.69 10299.20
```

```r
# R less than 1.1 indicates convergence

gelman.diag(samples)
```

```
## Potential scale reduction factors:
##
##      Point est. Upper C.I.
## mu.X          1          1
## mu.Y          1          1
##
## Multivariate psrf
##
## 1
```

```r
# |z| less than 2 indicates convergence

geweke.diag(samples[[1]])
```

```
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
##   mu.X   mu.Y
## -1.662 -1.146
```

MCMC failure example

Arnab Hazra

2/27/2022

We first simulate two sets of observations $X_1, \ldots, X_n \sim \text{Normal}(\mu_X, 1)$ and $Y_1, \ldots, Y_n \sim \text{Normal}(\mu_Y, 1)$. However, suppose these data are not observable and we only know $Z_i = X_i + Y_i, i = 1, \ldots, n$. Thus, $Z_i \sim \text{Normal}(\mu_X + \mu_Y, 2)$. We want to estimate μ_X and μ_Y. We choose priors $\mu_X, \mu_Y \sim \text{Normal}(0, 100^2)$.

```
rm(list = ls())

n <- 100

mu.X <- 2
mu.Y <- 4

set.seed(100)

X <- rnorm(n, mean = mu.X, sd = 1)
Y <- rnorm(n, mean = mu.Y, sd = 1)
Z <- X + Y

data <- list(Z = Z, n = n)
```

Next, we write the script for JAGS.

(1) Define the model as a string.

```
library(rjags)
```

```
## Warning: package 'rjags' was built under R version 4.0.5

## Loading required package: coda

## Linked to JAGS 4.3.0

## Loaded modules: basemod,bugs
```

```
model_string <- textConnection("model{

  # Likelihood (dnorm uses a precision, not variance)
  for(i in 1:n){
  Z[i] ~ dnorm(mu.X + mu.Y, 1 / 2)
  }

  # Priors

  mu.X ~ dnorm(0, 0.0001)
  mu.Y ~ dnorm(0, 0.0001)
  }")
```

(2) Load the data and compile the MCMC code

```r
inits <- list(mu.X = mean(Z)/2, mu.Y = mean(Z)/2)
model <- jags.model(model_string, data = data, inits = inits, quiet = TRUE, n.chains = 2)
```

(3) Burn-in for 1000 samples

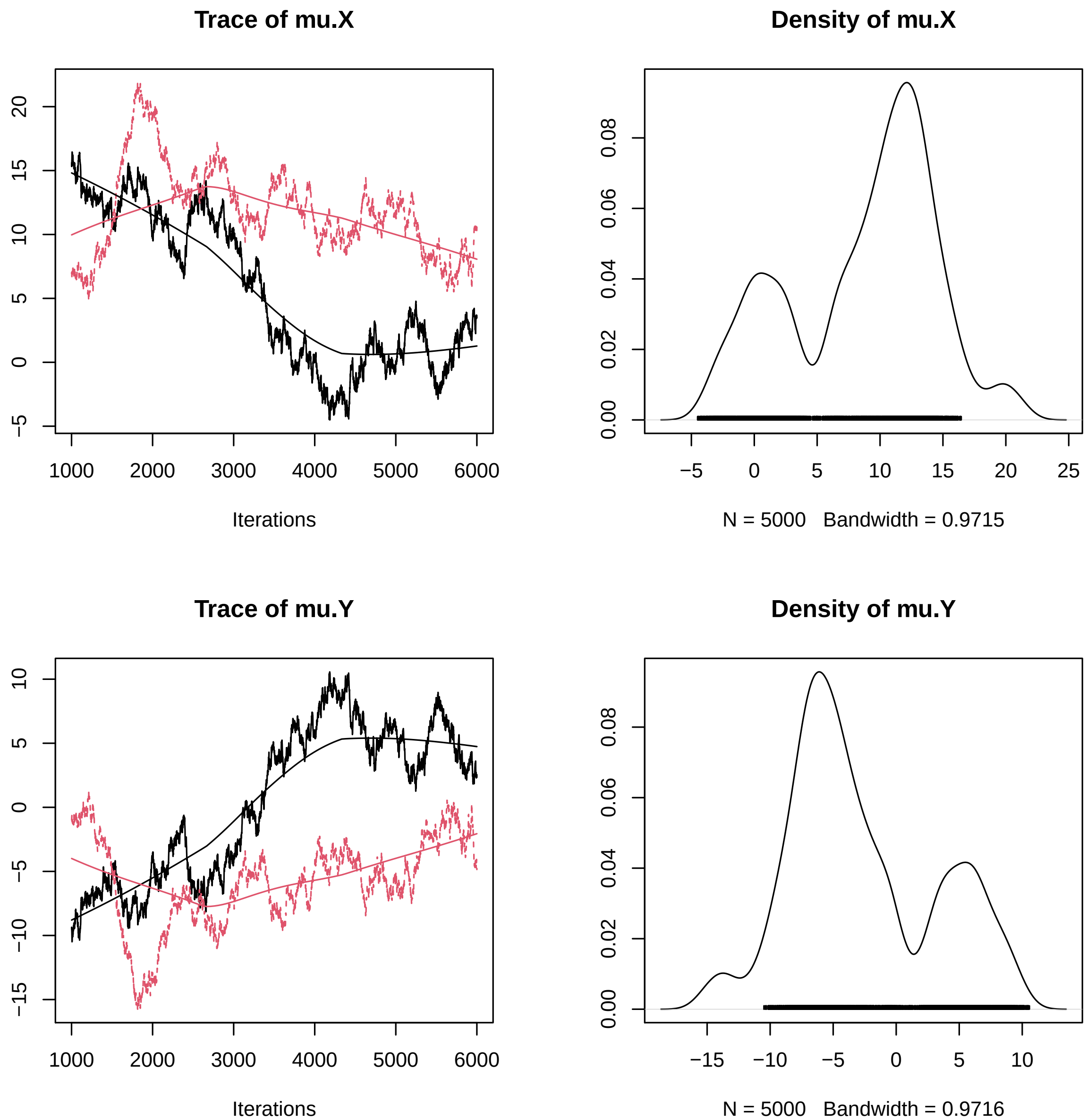
```r
update(model, 1000, progress.bar = "none")
```

(4) Generate 5000 post-burn-in samples

```r
params <- c("mu.X", "mu.Y")
samples <- coda.samples(model,
                        variable.names = params,
                        n.iter = 5000, progress.bar = "none")
```

(5) Graphical diagnostics

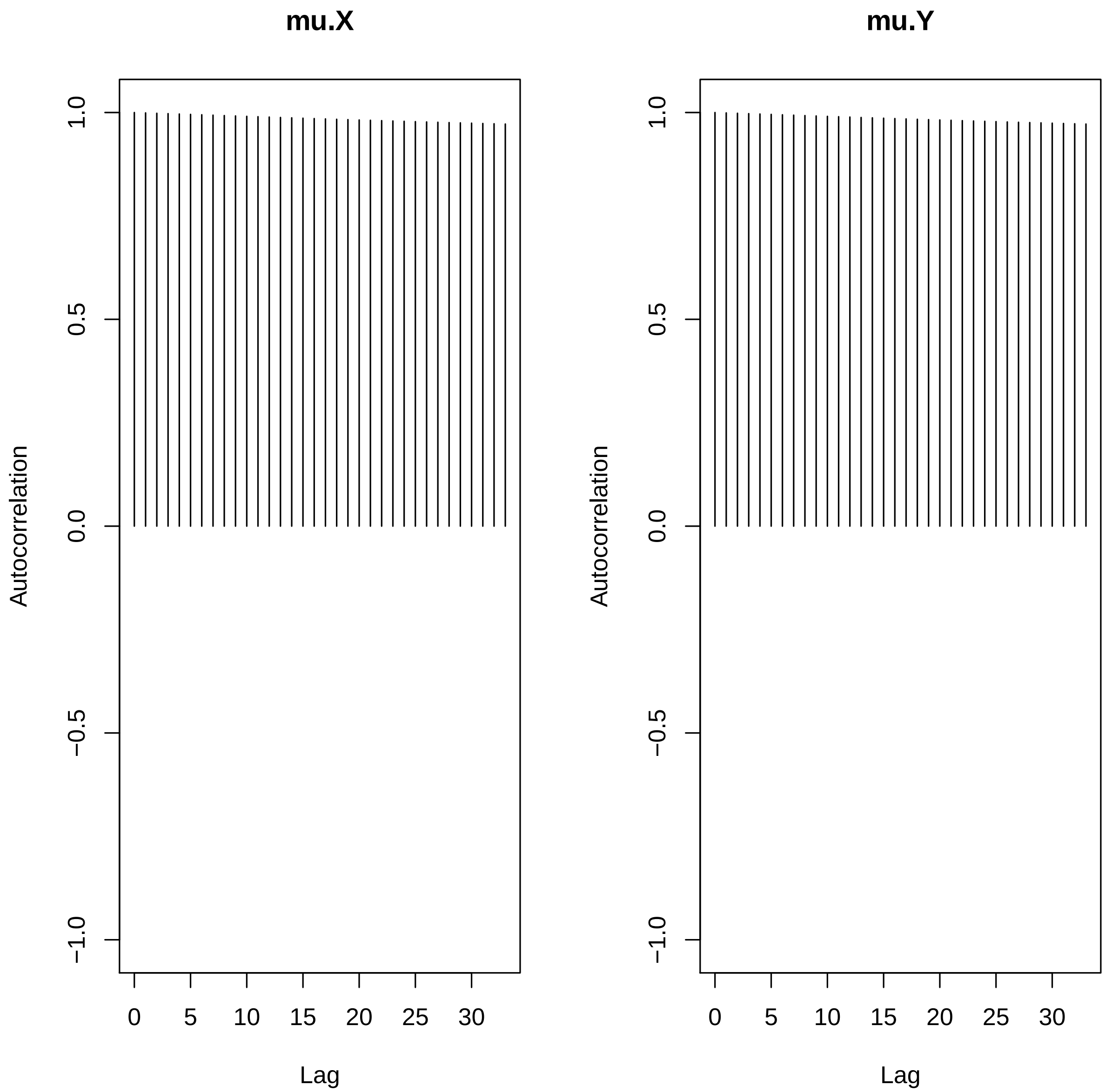
```r
plot(samples)
```

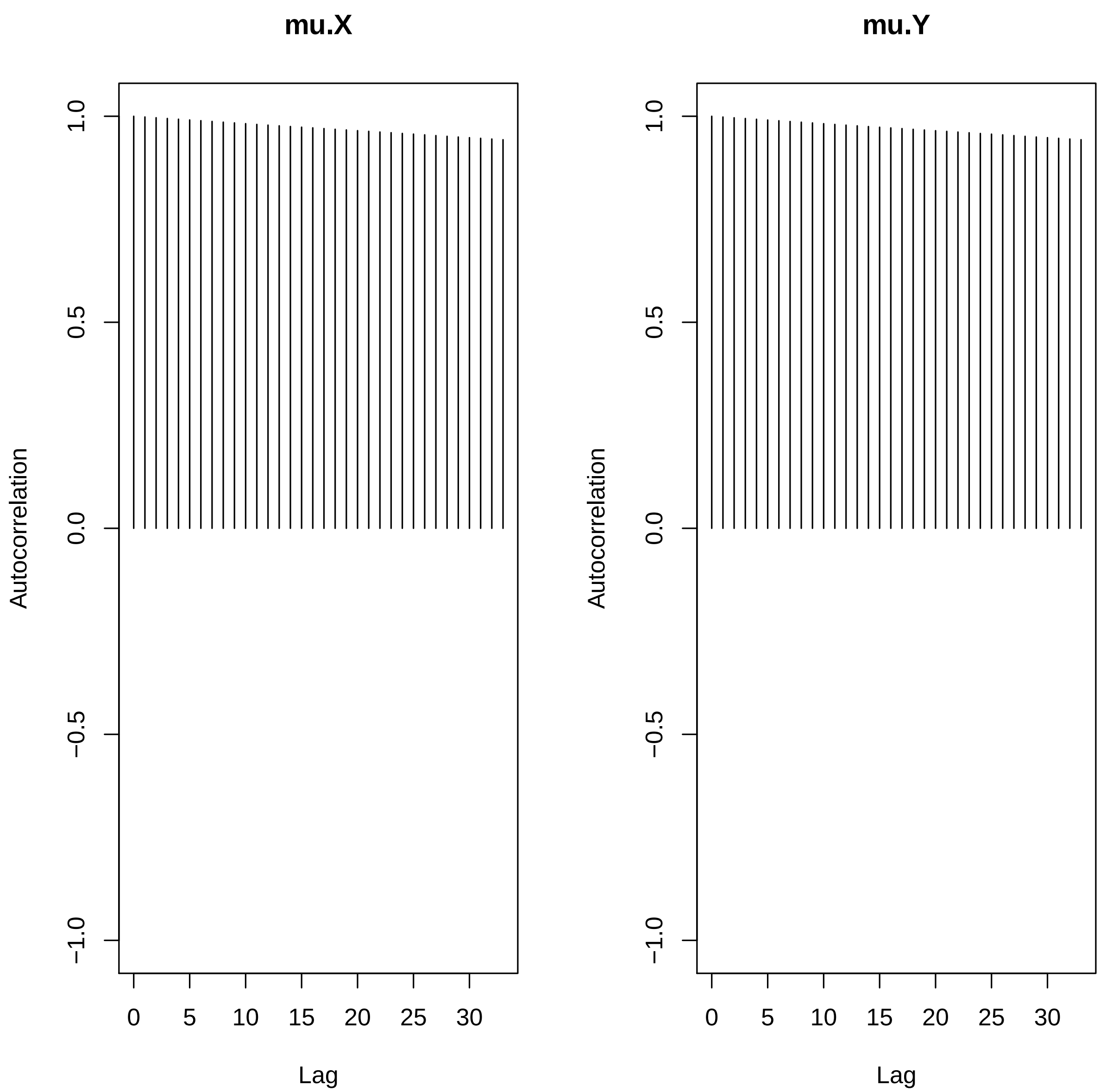


The trace plots do not look like caterpillars, the chains (one in black and one in gray) do not give similar estimates.

```r
library(coda)

autocorr.plot(samples)
```


mu.X
1.0
0.5
0.0
−0.5
−1.0
Autocorrelation
0
5
10
15
20
25
30
Lag
mu.Y
1.0
0.5
0.0
−0.5
−1.0
Autocorrelation
0
5
10
15
20
25
30
Lag



Autocorrelation is near zero for all lags. Together, these indicate convergence.

Numerical diagnostics

```
# Low autocorrelation indicates convergence

autocorr(samples[[1]], lag = 1)
```

```
## , , mu.X
##
##              mu.X      mu.Y
## Lag 1 0.9990959 -0.998801
##
## , , mu.Y
##
```

```
##            mu.X      mu.Y
## Lag 1 -0.999394 0.9991072
```

```r
# high ESS indicates convergence

effectiveSize(samples)
```

```
##     mu.X     mu.Y
## 6.684242 6.872405
```

```r
# R less than 1.1 indicates convergence

gelman.diag(samples)
```

```
## Potential scale reduction factors:
##
##      Point est. Upper C.I.
## mu.X       4.55       10.7
## mu.Y       4.55       10.7
##
## Multivariate psrf
##
## 3.3
```

```r
# |z| less than 2 indicates convergence

geweke.diag(samples[[1]])
```

```
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
##   mu.X   mu.Y
##  13.15 -13.17
```

Lecture 19

JAGS (software) essentials

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Options for coding MCMC

- Writing your own code.
- R packages for specific models (BLR for Bayesian Linear Regression, for example).
- All-purpose software like JAGS, BUGS, and STAN.

Why Just Another Gibbs Sampler (JAGS)?

- You can fit virtually any model.
- You can call `JAGS` from `R` which allows for plotting and data manipulation in `R`.
- It runs on all platforms: `LINUX`, `Mac`, `Windows`.
- There is a lot of help online.
- R has in-built packages for convergence diagnostics.

How does JAGS work?

- You specify the model by declaring the likelihood and priors.
- JAGS then sets up the MCMC sampler, e.g., works out the full conditional distributions for all parameters.
- It returns MCMC samples in a matrix or array.
- It also automatically produces posterior summaries like means, credible sets, and convergence diagnostics.
- User's manual: `https://people.stat.sc.edu/hansont/stat740/jags_user_manual.pdf`

Running JAGS from `R` has the following steps

1. Install JAGS: `https://sourceforge.net/projects/mcmc-jags/files/JAGS/4.x/Windows/`
2. Download `rjags` from CRAN and load the library.
3. Specify the model as a string.
4. Compile the model using the function `jags.model`.
5. Draw burn-in samples using the function `update`.
6. Draw posterior samples using the function `coda.samples`.
7. Inspect the results using the `plot` and `summary` functions.

Examples

- For moderately-sized problems `JAGS` is competitive with these methods.
- For really big and/or complex analyses `STAN` is preferred.
- `JAGS` is easier to code and so we will use it through the course, but you should be familiar with other software.
- Once you understand `JAGS`, switching to the others is straightforward.

Thank you!

JAGS illustration

Arnab Hazra

3/13/2023

First, suppose we simulate a dataset from the model $Y_i \sim \text{Normal}(\mu, \sigma^2)$. Then, suppose we choose priors $\mu \sim \text{Normal}(0, 100^2)$ and $\sigma^2 \sim \text{Inverse-Gamma}(0.01, 0.01)$.

```
rm(list = ls())

X <- rnorm(100, mean = 5, sd = 1)
```

Now, let us fit the same model back using JAGS.

(1) Define the model as a string.

```
library(rjags)
```

```
## Loading required package: coda
```

```
## Linked to JAGS 4.3.0
```

```
## Loaded modules: basemod,bugs
```

```
data <- list(X = X, n = length(X))

model_string <- textConnection("model{

   # Likelihood (dnorm uses a precision, not variance)
   for(i in 1:n){
     X[i] ~ dnorm(mu,tau) #tau = 1/sigma^2
   }

   # Priors
   tau ~ dgamma(0.01, 0.01)
   sigma <- 1/sqrt(tau)
   mu ~ dnorm(0, 0.0001)

 }")
```

(2) Load the data and compile the MCMC code

```
inits <- list(mu = mean(X), tau = 1 / var(X))
model <- jags.model(model_string, data = data, inits = inits, quiet = TRUE)
```

(3) Burn-in for 10000 samples

```
update(model, 10000, progress.bar = "none")
```

(4) Generate 20000 post-burn-in samples and retain the parameters named in params

```
params  <- c("mu", "sigma")
samples <- coda.samples(model,
```

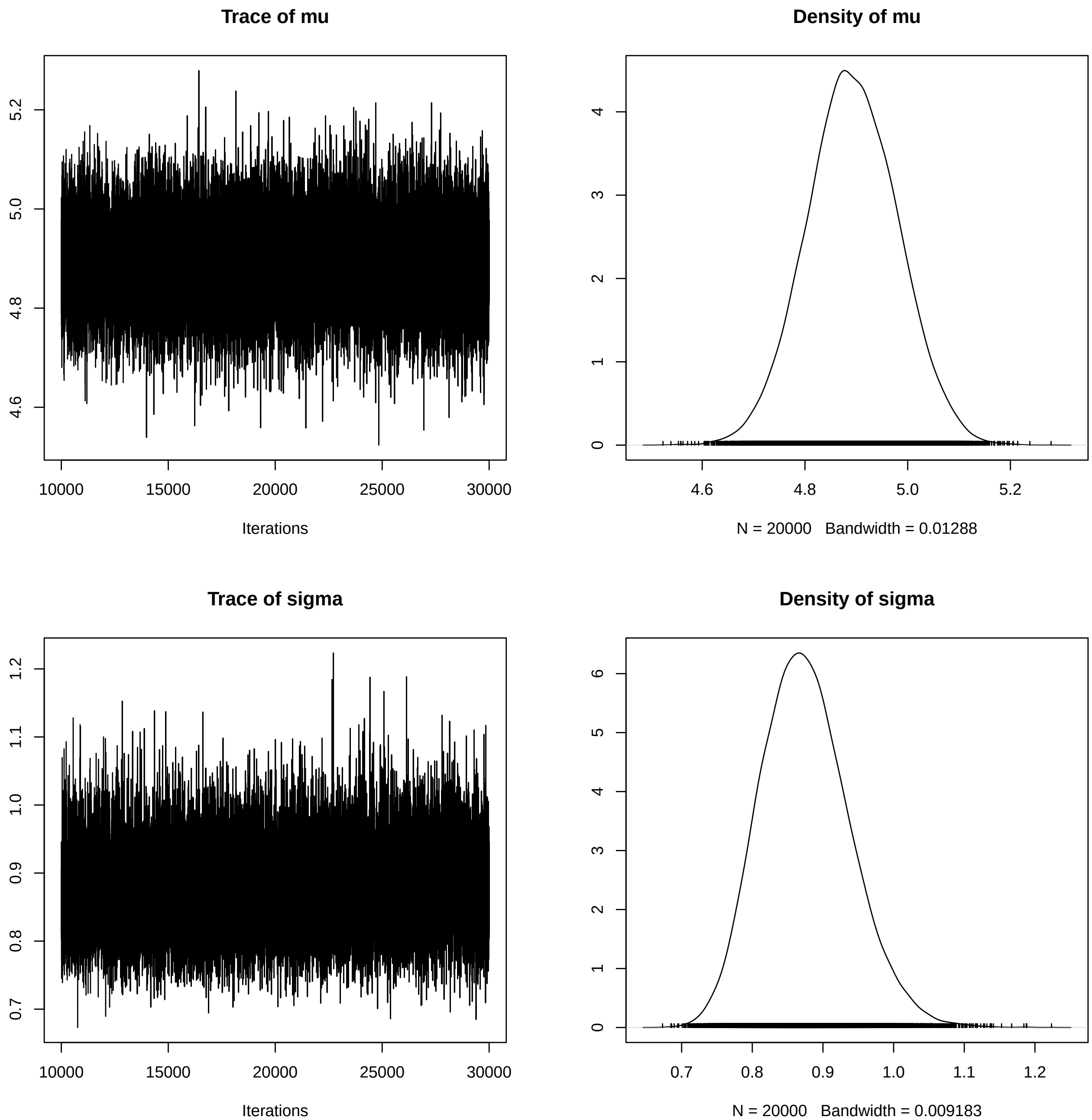
```
            variable.names = params,
            n.iter = 20000, progress.bar = "none")
```

(5) Summarize the output

```
summary(samples)
```

```
##
## Iterations = 10001:30000
## Thinning interval = 1
## Number of chains = 1
## Sample size per chain = 20000
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##         Mean      SD  Naive SE Time-series SE
## mu    4.8927 0.08809 0.0006229      0.0006229
## sigma 0.8753 0.06279 0.0004440      0.0004440
##
## 2. Quantiles for each variable:
##
##         2.5%    25%    50%    75% 97.5%
## mu    4.7214 4.8335 4.8917 4.9527 5.066
## sigma 0.7636 0.8311 0.8717 0.9154 1.008
```

```
plot(samples)
```


Trace of mu
Density of mu
Iterations
N = 20000   Bandwidth = 0.01288
Trace of sigma
Density of sigma
Iterations
N = 20000   Bandwidth = 0.009183

JAGS illustration

Arnab Hazra

3/13/2022

First, suppose we simulate a dataset from the model $Y_i \sim \text{Normal}(\beta_0 + \beta_1 X_i, \sigma^2)$. Then, suppose we choose priors $\beta_0, \beta_1 \sim \text{Normal}(0, 100^2)$ IID and $\sigma^2 \sim \text{Inverse-Gamma}(0.01, 0.01)$.

```
rm(list = ls())

X <- runif(500)

Y <- rnorm(500, mean = 5 + 2 * X, sd = 1)
```

Now, let us fit the same model back using JAGS.

(1) Define the model as a string.

```
library(rjags)
```

```
## Loading required package: coda
## Linked to JAGS 4.3.0
## Loaded modules: basemod,bugs
```

```
data <- list(Y =Y, X = X, n = length(X))

model_string <- textConnection("model{

   # Likelihood (dnorm uses a precision, not variance)
   for(i in 1:n){
     Y[i] ~ dnorm(beta0 + beta1 * X[i], tau) #tau = 1/sigma^2
   }

   # Priors
   tau ~ dgamma(0.01, 0.01)
   sigma <- 1/sqrt(tau)
   beta0 ~ dnorm(0, 0.0001)
   beta1 ~ dnorm(0, 0.0001)
 }")
```

(2) Load the data and compile the MCMC code

```
inits <- list(beta0 = mean(X), beta1 = 0, tau = 1 / var(X))
model <- jags.model(model_string, data = data, inits = inits, quiet = TRUE)
```

(3) Burn-in for 10000 samples

```
update(model, 10000, progress.bar = "none")
```

(4) Generate 20000 post-burn-in samples and retain the parameters named in params

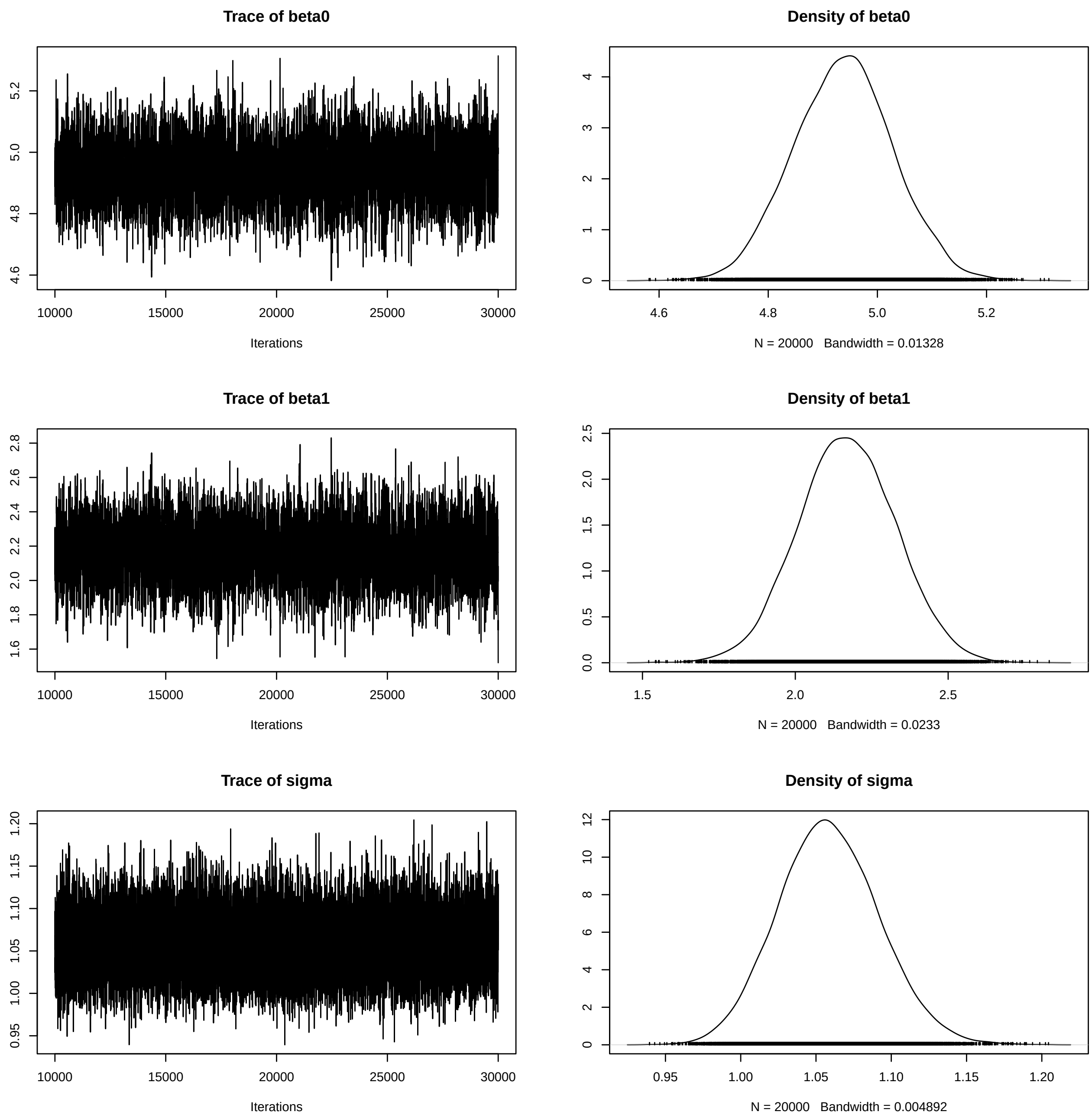
```r
params  <- c("beta0", "beta1", "sigma")
samples <- coda.samples(model,
            variable.names = params,
            n.iter = 20000, progress.bar = "none")
```

(5) Summarize the output

```r
summary(samples)
```

```
##
## Iterations = 10001:30000
## Thinning interval = 1
## Number of chains = 1
## Sample size per chain = 20000
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##        Mean      SD  Naive SE Time-series SE
## beta0 4.938 0.09079 0.0006420      0.0016109
## beta1 2.169 0.15934 0.0011267      0.0028318
## sigma 1.059 0.03345 0.0002365      0.0002365
##
## 2. Quantiles for each variable:
##
##         2.5%   25%   50%   75% 97.5%
## beta0 4.7619 4.876 4.939 4.998 5.115
## beta1 1.8578 2.062 2.168 2.276 2.480
## sigma 0.9952 1.036 1.058 1.081 1.126
```

```r
plot(samples)
```


Trace of beta0
Iterations
Density of beta0
N = 20000   Bandwidth = 0.01328
Trace of beta1
Iterations
Density of beta1
N = 20000   Bandwidth = 0.0233
Trace of sigma
Iterations
Density of sigma
N = 20000   Bandwidth = 0.004892

JAGS illustration

Arnab Hazra

3/13/2022

Now, suppose we consider a random effect model.

We simulate a dataset from the model $Y_i|X_i \sim \text{Normal}(X_i, \sigma^2)$ and $X_i \sim \text{Normal}(\mu, 1)$. Thus, the model is actually $Y_i \sim \text{Normal}(\mu, 1+\sigma^2)$ Then, suppose we choose priors $\mu \sim \text{Normal}(0, 100^2)$ and $\sigma^2 \sim \text{Inverse-Gamma}(0.01, 0.01)$.

```
rm(list = ls())

mu <- 5
sigma <- 1

X <- rnorm(500, mean = mu, sd = 1)

Y <- rnorm(500, mean = X, sd = sigma)
```

Now, let us fit the same model back using JAGS.

(1) Define the model as a string.

```
library(rjags)
```

```
## Loading required package: coda
```

```
## Linked to JAGS 4.3.0
```

```
## Loaded modules: basemod,bugs
```

```
data <- list(Y = Y, n = length(Y))

model_string <- textConnection("model{

   # Likelihood (dnorm uses a precision, not variance)
   for(i in 1:n){
     Y[i] ~ dnorm(X[i], tau) #tau = 1/sigma^2
   }

   # Priors
   for(i in 1:n){
     X[i] ~ dnorm(mu, 1)
   }
   tau ~ dgamma(0.01, 0.01)
   sigma <- 1/sqrt(tau)
   mu ~ dnorm(0, 0.0001)
 }")
```

(2) Load the data and compile the MCMC code

```r
inits <- list(mu = mean(Y), tau = 1 / var(Y))
model <- jags.model(model_string, data = data, inits = inits, quiet = TRUE)
```

(3) Burn-in for 10000 samples

```r
update(model, 10000, progress.bar = "none")
```

(4) Generate 20000 post-burn-in samples and retain the parameters named in params

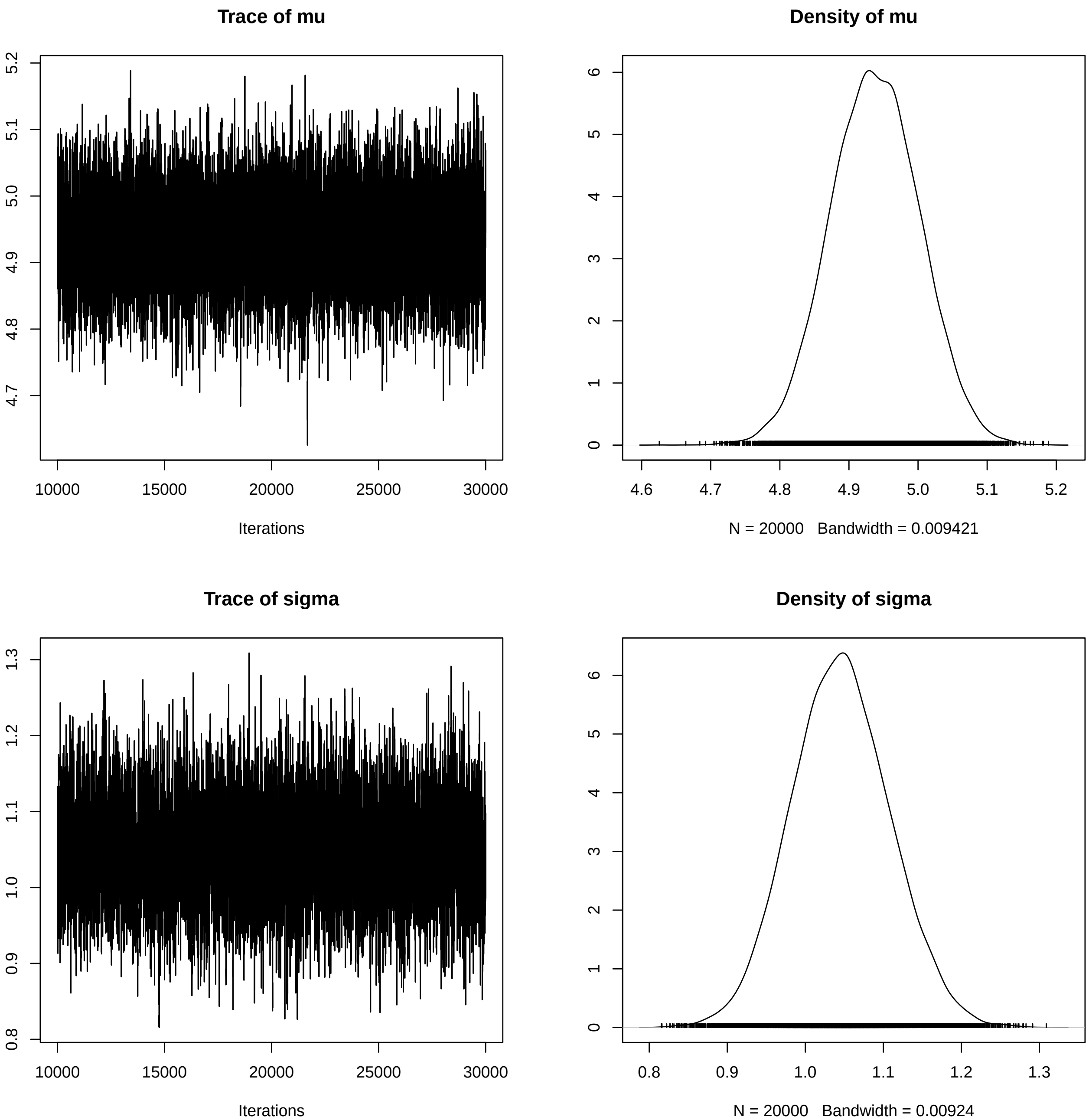
```r
params  <- c("mu", "sigma")
samples <- coda.samples(model,
            variable.names = params,
            n.iter = 20000, progress.bar = "none")
```

(5) Summarize the output

```r
summary(samples)
```

```
##
## Iterations = 10001:30000
## Thinning interval = 1
## Number of chains = 1
## Sample size per chain = 20000
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##        Mean      SD  Naive SE Time-series SE
## mu    4.938 0.06441 0.0004555      0.0008166
## sigma 1.046 0.06318 0.0004467      0.0011215
##
## 2. Quantiles for each variable:
##
##         2.5%   25%   50%   75% 97.5%
## mu    4.8122 4.894 4.938 4.982 5.063
## sigma 0.9248 1.003 1.045 1.088 1.171
```

```r
plot(samples)
```


Trace of mu
Iterations
Density of mu
N = 20000 Bandwidth = 0.009421
Trace of sigma
Iterations
Density of sigma
N = 20000 Bandwidth = 0.00924

Lecture 19

Bayesian Linear Models Part 1

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Linear regression

- Linear regression is by far the most common statistical model.
- It includes as special cases the t-test and ANOVA.
- The multiple linear regression model is

  $$Y_i \sim \text{Normal}(\beta_0 + X_{i1}\beta_1 + ... + X_{ip}\beta_p, \sigma^2),$$

  independently across the $i = 1, ..., n$ observations.
- As we'll see, Bayesian and classical linear regression are similar if $n >> p$ and the priors are non-informative.
- However, the results can be different for challenging problems, and the interpretation is different in all cases.

Outline of Chapter 4

- Bayesian t-tests
- Bayesian linear regression
  - Gaussian priors
  - Jeffreys' priors
  - Shrinkage priors
- Generalized linear models
- Random effects
- Flexible linear models
  - Non-linear regression
  - Heteroskedastic errors
  - Non-Gaussian errors
  - Correlated errors

Bayesian one-sample (i.e., paired) t-test

- Say $Y_1, ..., Y_n \sim \text{Normal}(\mu, \sigma^2)$.
- Typically Y_i is the difference of a pair of measurements, e.g., the post- minus pre-test for subject i.
- Therefore the interest is to compare μ to zero.
- We will consider two cases: σ^2 known and σ^2 unknown.

Jeffreys priors for the model $Y \sim \text{Normal}(\mu, 1)$

- The log likelihood is

$$\ell(\theta) = \log[f(Y|\theta)] = -\frac{1}{2}\log(2\pi) - \frac{1}{2}(Y-\mu)^2.$$

- The first derivative is

$$\ell'(\theta) = Y - \mu.$$

- The second derivative is

$$\ell''(\theta) = -1.$$

- Therefore,

$$I(\theta) = -E(\ell''(\theta)) = 1.$$

- Finally, the prior is

$$\pi(\theta) \propto I(\theta)^{1/2} = 1.$$

Bayesian one-sample (i.e., paired) t-test

- For fixed σ, the Jeffreys' prior for μ is $\pi(\mu) = 1$.
- The posterior is

$$
\begin{aligned}
p(\mu|\mathbf{Y}) &\propto \exp\left[-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(Y_i - \mu)^2\right] \\
&\propto \exp\left[-\frac{n\mu^2}{2\sigma^2}\right] \times \exp\left[\frac{\mu\sum_{i=1}^{n} Y_i}{\sigma^2}\right] \\
&\propto \exp\left[-\frac{n}{2}(\mu - \bar{Y})^2\right].
\end{aligned}
$$

By matching the terms with those in the density of a normal distribution, we have

$$\mu|\mathbf{Y}, \sigma \sim \text{Normal}\left(\bar{Y}, \frac{\sigma^2}{n}\right).$$

Bayesian one-sample (i.e., paired) t-test

- Therefore the posterior mean is the sample mean,

$$E(\mu|\mathbf{Y}) = \bar{Y}.$$

- The 95% credible set is

$$\bar{Y} \pm 1.96\frac{\sigma}{\sqrt{n}}.$$

- For the test of $\mathcal{H}_0 : \mu \leq 0$ versus $\mathcal{H}_1 : \mu > 0$,

$$\text{Prob}(\mathcal{H}_0|\mathbf{Y}) = \text{Prob}(\mu \leq 0|\mathbf{Y}) = \Phi(\sqrt{n}\bar{Y}/\sigma)$$

is the frequentist p-value.

Jeffreys priors for σ for the model $Y \sim \text{Normal}(0, \sigma^2)$

- The log likelihood is

$$\ell(\theta) = \log[f(Y|\theta)] = -\frac{1}{2}\log(2\pi) - \log(\sigma) - \frac{1}{2}\frac{Y^2}{\sigma^2}.$$

- The first derivative is

$$\ell'(\theta) = -\frac{1}{\sigma} + \frac{Y^2}{\sigma^3}.$$

- The second derivative is

$$\ell''(\theta) = \frac{1}{\sigma^2} - 3\frac{Y^2}{\sigma^4}.$$

- Therefore,

$$I(\theta) = -E(\ell''(\theta)) = -\frac{1}{\sigma^2} + 3\frac{\sigma^2}{\sigma^4} = \frac{2}{\sigma^2}.$$

- Finally, the prior is

$$\pi(\theta) \propto I(\theta)^{1/2} \propto \frac{1}{\sigma}.$$

Jeffreys priors for σ^2 for the model $Y \sim \text{Normal}(0, \sigma^2)$

- The log likelihood is

$$\ell(\theta) = \log[f(Y|\theta)] = -\frac{1}{2}\log(2\pi) - \frac{1}{2}\log(\sigma^2) - \frac{1}{2}\frac{Y^2}{\sigma^2}.$$

- The first derivative is

$$\ell'(\theta) = -\frac{1}{2}\frac{1}{\sigma^2} + \frac{1}{2}\frac{Y^2}{(\sigma^2)^2}.$$

- The second derivative is

$$\ell''(\theta) = \frac{1}{2}\frac{1}{(\sigma^2)^2} - \frac{Y^2}{(\sigma^2)^3}.$$

- Therefore,

$$I(\theta) = -E(\ell''(\theta)) = -\frac{1}{2}\frac{1}{\sigma^4} + \frac{\sigma^2}{\sigma^6} = \frac{1}{2}\frac{1}{\sigma^4}.$$

- Finally, the prior is

$$\pi(\theta) \propto I(\theta)^{1/2} \propto \frac{1}{\sigma^2}.$$

Jeffreys priors for $\boldsymbol{\theta} = (\mu, \sigma^2)'$ for the model $Y \sim \text{Normal}(\mu, \sigma^2)$

- The log likelihood is

$$\ell(\boldsymbol{\theta}) = \log[f(Y|\boldsymbol{\theta})] = -\frac{1}{2}\log(2\pi) - \frac{1}{2}\log(\sigma^2) - \frac{1}{2}\frac{(Y-\mu)^2}{\sigma^2}.$$

- The first derivative is

$$\ell_1'(\boldsymbol{\theta}) = \frac{Y-\mu}{\sigma^2}, \quad \ell_2'(\boldsymbol{\theta}) = -\frac{1}{2}\frac{1}{\sigma^2} + \frac{1}{2}\frac{(Y-\mu)^2}{(\sigma^2)^2}$$

- The second derivative is

$$\ell_{11}''(\boldsymbol{\theta}) = -\frac{1}{\sigma^2}, \;\; \ell_{12}''(\boldsymbol{\theta}) = \ell_{21}''(\boldsymbol{\theta}) = -\frac{Y-\mu}{(\sigma^2)^2}, \;\; \ell_{22}''(\boldsymbol{\theta}) = \frac{1}{2}\frac{1}{(\sigma^2)^2} - \frac{(Y-\mu)^2}{(\sigma^2)^3}.$$

Jeffreys priors for $\boldsymbol{\theta} = (\mu, \sigma^2)'$ for the model $Y \sim$ Normal(μ, σ^2)

- Therefore,

$$
\begin{aligned}
-E(\ell''_{11}(\boldsymbol{\theta})) &= \frac{1}{\sigma^2} \\
-E(\ell''_{12}(\boldsymbol{\theta})) &= -E(\ell''_{21}(\theta)) = 0 \\
-E(\ell''_{22}(\boldsymbol{\theta})) &= \frac{1}{2}\frac{1}{\sigma^4}.
\end{aligned}
$$

- Finally, the prior is

$$
\pi(\boldsymbol{\theta}) \propto |I(\boldsymbol{\theta})|^{1/2} \propto \frac{1}{\sigma^3}.
$$

Bayesian one-sample (i.e., paired) t-test

- When σ^2 is unknown, the Jeffreys' prior is

$$\pi(\mu, \sigma^2) \propto \left(\frac{1}{\sigma^2}\right)^{3/2}.$$

- The marginal posterior integrating over uncertainty in σ^2 is

$$\mu|\mathbf{Y} \sim t_n\left(\bar{Y}, \frac{\hat{\sigma}^2}{n}\right),$$

where $\hat{\sigma}^2 = \sum_{i=1}^n (Y_i - \bar{Y})^2/n$.

- This is very similar to the frequentist t-test, except that the degrees of freedom is n rather than $n-1$.
- This is the effect of the prior.

Bayesian two-sample t-test

- Say the n_1 observations from group 1 are

$$Y_i \sim \text{Normal}(\mu, \sigma^2),$$

  and the n_2 observations from group 2 are

$$Y_i \sim \text{Normal}(\mu + \delta, \sigma^2).$$

- The goal is to compare δ to zero.
- With σ^2 known and Jeffrey's prior $\pi(\mu, \delta) = 1$,

$$\delta | \mathbf{Y}, \sigma^2 \sim \text{Normal}\left(\bar{Y}_2 - \bar{Y}_1, \frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}\right),$$

  and the results are identical to the two-sample z-test.

Bayesian two-sample t-test

- When σ^2 is unknown, the Jeffreys' prior is
$$\pi(\mu, \delta, \sigma^2) \propto \left(\frac{1}{\sigma^2}\right)^2.$$
- The marginal posterior integrating over uncertainty in σ^2 and μ is
$$\delta|\mathbf{Y} \sim t_n\left(\bar{Y}_2 - \bar{Y}_1, \frac{\hat{\sigma}^2}{n_1} + \frac{\hat{\sigma}^2}{n_2}\right),$$
where the pooled variance estimator is
$$\hat{\sigma}^2 = \left[\sum_{i=1}^{n_1}(Y_i - \bar{Y}_1)^2 + \sum_{i=n_1+1}^{n_2}(Y_i - \bar{Y}_2)^2\right]/n.$$
- This is very similar to the frequentist t-test, except that the degrees of freedom is $n = n_1 + n_2$ rather than $n - 2$.
- This is the effect of the prior.

Review of least squares

- The least squares estimate of $\boldsymbol{\beta} = (\beta_0, \beta_1, ..., \beta_p)^T$ is

$$\hat{\boldsymbol{\beta}}_{OLS} = \underset{\boldsymbol{\beta}}{\text{argmin}} \sum_{i=1}^{n} (Y_i - \mu_i)^2,$$

  where $\mu_i = \beta_0 + X_{i1}\beta_1 + ... + X_{ip}\beta_p$.

- $\hat{\boldsymbol{\beta}}_{OLS}$ is unbiased even if the errors are non-Gaussian.
- If the errors are Gaussian, then the likelihood is proportional to

$$\prod_{i=1}^{n} \exp\left[-\frac{(Y_i - \mu_i)^2}{2\sigma^2}\right] = \exp\left[-\frac{\sum_{i=1}^{n}(Y_i - \mu_i)^2}{2\sigma^2}\right].$$

- Therefore, if the errors are Gaussian $\hat{\boldsymbol{\beta}}_{OLS}$ is also the MLE.

Review of least squares

- Linear regression is often simpler to describe using linear algebra notation.
- Let $\mathbf{Y} = (Y_1, ..., Y_n)^T$ be the response vector and $\mathbf{X}$ be the $n \times (p+1)$ matrix of covariates.
- Then the mean of $\mathbf{Y}$ is $\mathbf{X}\beta$ and the least squares solution is

$$\hat{\beta}_{OLS} = \underset{\beta}{\text{argmin}}(\mathbf{Y} - \mathbf{X}\beta)^T(\mathbf{Y} - \mathbf{X}\beta) = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}.$$

- If the errors are Gaussian then the sampling distribution is

$$\hat{\beta}_{OLS} \sim \text{Normal}\left[\beta, \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}\right].$$

- If the variance σ^2 is estimated using the mean squared residual error, then the sampling distribution is multivariate t.

Bayesian regression

- The likelihood remains

$$Y_i \sim \text{Normal}(\beta_0 + X_{i1}\beta_1 + ... + X_{ip}\beta_p, \sigma^2)$$

  independent for $i = 1, ..., n$ observations

- As with a least squares analysis, it is crucial to verify this is appropriate using qq-plots, added variable plots, etc.

- A Bayesian analysis also requires priors for β and σ

- We will focus on prior specification since this piece is uniquely Bayesian.

Thank you!

Lecture 20

Bayesian Linear Models Part 2

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Recap: Linear regression

- Linear regression is by far the most common statistical model.
- It includes as special cases the t-test and ANOVA.
- The multiple linear regression model is

$$Y_i \sim \text{Normal}(\beta_0 + X_{i1}\beta_1 + ... + X_{ip}\beta_p, \sigma^2),$$

  independently across the $i = 1, ..., n$ observations.
- As we'll see, Bayesian and classical linear regression are similar if $n >> p$ and the priors are non-informative.
- However, the results can be different for challenging problems, and the interpretation is different in all cases.

Recap: Review of least squares

- The least squares estimate of $\boldsymbol{\beta} = (\beta_0, \beta_1, ..., \beta_p)^T$ is

$$\hat{\boldsymbol{\beta}}_{OLS} = \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \sum_{i=1}^{n} (Y_i - \mu_i)^2,$$

  where $\mu_i = \beta_0 + X_{i1}\beta_1 + ... + X_{ip}\beta_p$.

- $\hat{\boldsymbol{\beta}}_{OLS}$ is unbiased even if the errors are non-Gaussian.
- If the errors are Gaussian, then the likelihood is proportional to

$$\prod_{i=1}^{n} \exp\left[-\frac{(Y_i - \mu_i)^2}{2\sigma^2}\right] = \exp\left[-\frac{\sum_{i=1}^{n}(Y_i - \mu_i)^2}{2\sigma^2}\right].$$

- Therefore, if the errors are Gaussian $\hat{\boldsymbol{\beta}}_{OLS}$ is also the MLE.

Recap: Review of least squares

- Linear regression is often simpler to describe using linear algebra notation.
- Let $\mathbf{Y} = (Y_1, ..., Y_n)^T$ be the response vector and $\mathbf{X}$ be the $n \times (p+1)$ matrix of covariates.
- Then the mean of $\mathbf{Y}$ is $\mathbf{X}\beta$ and the least squares solution is

$$\hat{\beta}_{OLS} = \underset{\beta}{\text{argmin}} (\mathbf{Y} - \mathbf{X}\beta)^T (\mathbf{Y} - \mathbf{X}\beta) = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}.$$

- If the errors are Gaussian then the sampling distribution is

$$\hat{\beta}_{OLS} \sim \text{Normal}\left[\beta, \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}\right].$$

- If the variance σ^2 is estimated using the mean squared residual error, then the sampling distribution is multivariate t.

Recap: Bayesian regression

- The likelihood remains

$$Y_i \sim \text{Normal}(\beta_0 + X_{i1}\beta_1 + ... + X_{ip}\beta_p, \sigma^2)$$

  independent for $i = 1, ..., n$ observations.

- As with a least squares analysis, it is crucial to verify this is appropriate using qq-plots, added variable plots, etc.

- A Bayesian analysis also requires priors for β and σ.

- We will focus on prior specification since this piece is uniquely Bayesian.

Priors

- For the purpose of setting priors, it is helpful to standardize both the response and each covariate to have mean zero and variance one.
- Many priors for β have been considered in the literature:
  1. Improper priors
  2. Gaussian priors
  3. Double exponential priors
  4. Many, many more...

Improper priors

- With σ fixed, the Jeffreys' prior is flat $p(\beta) = 1$.
- This is improper, but the posterior is proper under the same conditions required by least squares.
- If σ is known then
$$\boldsymbol{\beta}|\mathbf{Y} \sim \text{Normal}\left[\hat{\boldsymbol{\beta}}_{OLS}, \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}\right].$$
- Therefore, the results should be similar to least squares.
- How are they different?

Improper priors

- Of course we rarely know σ.
- A conjugate uninformative prior is

$$\sigma^2 \sim \text{InvGamma}(a, b),$$

  with a and b set to be small, say $a = b = 0.01$.
- In this case, the posterior of β follows a multivariate t centered on $\hat{\beta}_{OLS}$.
- Again, the results are similar to OLS.

Improper priors

- The objective Bayes Jeffreys prior is

$$p(\beta, \sigma^2) = \left(\frac{1}{\sigma^2}\right)^{p/2+1},$$

which is the inverse gamma prior with $a = p/2$ and $b \to 0$.

- This gives posterior (marginal over σ^2)

$$\beta|\mathbf{Y} \sim \mathrm{t}_n\left(\hat{\beta}_{OLS}, \hat{\sigma}^2(\mathbf{X}^T\mathbf{X})^{-1}\right),$$

where $\hat{\sigma}^2 = (\mathbf{Y} - \mathbf{X}\hat{\beta}_{OLS})^T(\mathbf{Y} - \mathbf{X}\hat{\beta}_{OLS})/n$.

- The posterior is proper in the same situations that the least squares solution exists.

Multivariate normal prior

- Another common prior for is Zellner's g-prior

$$\beta \sim \text{Normal}\left[0, \frac{\sigma^2}{g}(\mathbf{X}^T\mathbf{X})^{-1}\right].$$

- This prior is proper assuming **X** is full rank.
- The posterior mean is

$$\frac{1}{1+g}\hat{\beta}_{OLS}.$$

- This shrinks the least estimate towards zero.
- g controls the amount of shrinkage.
- $g = 1/n$ is common, and called the unit information prior.

Univariate Gaussian priors

- If there are many covariates or the covariates are collinear, then $\hat{\beta}_{OLS}$ is unstable.
- Independent priors can counteract collinearity

$$\beta_j \sim \text{Normal}(0, \sigma^2/g)$$

  independent over j.
- The posterior mode is

$$\underset{\beta}{\text{argmin}}\{\sum_{i=1}^{n}(Y_i - \mu_i)^2 + g\sum_{j=1}^{p}\beta_j^2\}.$$

- In classical statistics, this is known as the ridge regression solution and is used to stabilize the least squares solution.

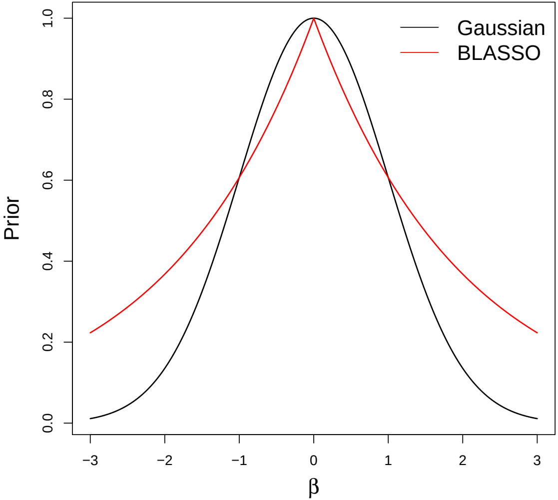
BLASSO

- An increasingly-popular prior is the double exponential or Bayesian LASSO prior.
- The prior is $\beta_j \sim \text{DE}(\tau)$ which has PDF

$$f(\beta) \propto \exp\left(-\frac{|\beta|}{\tau}\right).$$

- The square in the Gaussian prior is replaced with an absolute value.
- The shape of the PDF is thus more peaked at zero (next slide).
- The BLASSO prior favors settings where there are many β_j near zero and a few large β_j.
- That is, p is large but most of the covariates are noise.

BLASSO



BLASSO

- The posterior mode is

$$\underset{\beta}{\operatorname{argmin}}\{\sum_{i=1}^{n}(Y_i - \mu_i)^2 + g\sum_{j=1}^{p}|\beta_j|\}$$

- In classical statistics, this is known as the LASSO solution.
- It is popular because it adds stability by shrinking estimates towards zero, and also sets some coefficients to zero.
- Covariates with coefficients set to zero can be removed.
- Therefore, LASSO performs variables selection and estimation simultaneously.

Computing

- With flat or Gaussian (with fixed prior variance) priors the posterior is available in closed-form and Monte Carlo sampling is not needed.
- JAGS also works well, but there are `R` packages dedicated just to Bayesian linear regression that are preferred for big/hard problems.
- `BLR` is probably the most common.

Computing for the BLASSO

- For the BLASSO prior, the full conditionals are more complicated.
- There is a trick to make all full conditionals conjugate so that Gibbs sampling can be used.
- Metropolis sampling works fine too.
- `BLR` works well for BLASSO and is super fast.

Summarizing the results

- The standard summary is a table with marginal means and 95% intervals for each β_j.
- This becomes unwieldy for large p.
- Picking a subset of covariates is a crucial step in a linear regression analysis.
- We will discuss this later in the course.
- Common methods include cross-validation, information criteria, and stochastic search.

Predictions

- Say we have a new covariate vector $\mathbf{X}_{new}$ and we would like to predict the corresponding response Y_{new}.
- A plug-in approach would fix β and σ at their posterior means $\hat{\beta}$ and $\hat{\sigma}$ to make predictions
$$Y_{new}|\hat{\beta}, \hat{\sigma} \sim \text{Normal}(\mathbf{X}_{new}\hat{\beta}, \hat{\sigma}^2).$$
- However this plug-in approach suppresses uncertainty about β and σ.
- Therefore these prediction intervals will be slightly too narrow leading to undercoverage.

Posterior predicitive distribution (PPD)

- We should really account for all uncertainty when making predictions, including our uncertainty about β and σ.
- We really want the PPD

$$
\begin{aligned}
p(Y_{new}|\mathbf{Y}) &= \int f(Y_{new}, \beta, \sigma|\mathbf{Y}) d\beta d\sigma \\
&= \int f(Y_{new}|\beta, \sigma) f(\beta, \sigma|\mathbf{Y}) d\beta d\sigma.
\end{aligned}
$$

- Marginalizing over the model parameters accounts for their uncertainty.
- The concept of the PPD applies generally (e.g., logistic regression) and means the distribution of the predicted value marginally over model parameters.

Posterior predicitive distribution (PPD)

- MCMC naturally gives draws from Y_{new}'s PPD
  - For MCMC iteration t we have $\beta^{(t)}$ and $\sigma^{(t)}$
  - For MCMC iteration t we sample
    $$Y_{new}^{(t)} \sim \text{Normal}(\mathbf{X}\beta^{(t)}, {\sigma^{(t)}}^2)$$
  - $Y_{new}^{(1)}, ..., Y_{new}^{(S)}$ are samples from the PPD
- This is an example of the claim that “Bayesian methods naturally quantify uncertainty”.

Thank you!

Lecture 21

Bayesian GLM and random effects models

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Generalized linear models

- Other forms of regression follow naturally from linear regression.
- For example, for binary responses $Y_i \in \{0, 1\}$ we might use logistic regression

$$\text{logit}[\text{Prob}(Y_i = 1)] = \eta_i = \beta_0 + \beta_1 X_{i1} + ... + \beta_p X_{ip}.$$

- The logit link is the log-odds $\text{logit}(x) = \log[x/(1-x)]$.
- Then β_j represents the increase in the log odds of an event corresponding to a one-unit increase in covariate j.
- The expit transformation $\text{expit}(x) = \exp(x)/[1 + \exp(x)]$ is the inverse, and

$$\text{Prob}(Y_i = 1) = \text{expit}(\eta_i) \in [0, 1].$$

Bayesian logistic regression

- Bayesian logistic regression requires a prior for β.
- All of the prior we have discussed for linear regression (Zellner, BLASSO, etc) are also applicable here.
- Computationally the full conditional distributions are no longer conjugate and so we must use Metropolis sampling.
- The `R` function `MCMClogit` does this efficiently.
- Other GLMs (e.g., Poisson regression, probit regression) are similar to implement using Bayesian methods.

Probit and Poisson regression

- For binary responses $Y_i \in \{0, 1\}$, we might use probit regression

$$\Phi^{-1}[\text{Prob}(Y_i = 1)] = \eta_i = \beta_0 + \beta_1 X_{i1} + ... + \beta_p X_{ip}.$$

- The probit link is $\text{probit}(x) = \Phi^{-1}(x)$, where Φ is Normal$(0, 1)$ CDF.

- For count responses $Y_i \in \{0, 1, \ldots\}$, we might use $Y_i \overset{Indep}{\sim}$ Poisson(λ_i) where

$$\log[\lambda_i] = \eta_i = \beta_0 + \beta_1 X_{i1} + ... + \beta_p X_{ip}.$$

- Computationally the full conditional distributions are no longer conjugate and so we must use Metropolis sampling.

Random effects

- Linear regression assumes that the errors are independent.
- This is invalid if data are grouped.
- For example, n classrooms each have m students.
- It might be reasonable to assume the classrooms are independent, but the students within a class are likely dependent.
- Choosing random effects is a natural way to account for this dependence.

One-way random effects model

- Say Y_{ij} is the score for student i in class j.
- The random effects model is

$$Y_{ij} = \alpha_j + \varepsilon_{ij}.$$

- The random effect for classroom j is α_j.
- This is viewed as a random draw from the population,

$$\alpha_j \overset{IID}{\sim} \text{Normal}(\mu, \tau^2).$$

- The population is described by μ and τ.
- The random errors are $\varepsilon_{ij} \overset{IID}{\sim} \text{Normal}(0, \sigma^2)$, independent over i and j.

One-way random effects model

- Conditioned on the classroom mean α_j all observations are independent:

$$Y_{ij}|\alpha_j \stackrel{Indep}{\sim} \text{Normal}(\alpha_j, \sigma^2).$$

- Marginalizing over the random effects gives

$$\text{Cor}(Y_{ij}, Y_{uv}) = \begin{cases} 0 & \text{for } j \neq v \\ \frac{\tau^2}{\sigma^2+\tau^2} & \text{for } j = v \end{cases}$$

- Therefore, in this model observations with the same classroom are correlated.

One-way random effects model

- To complete the Bayesian model, we must specify priors for μ, σ^2 and τ.
- A normal prior with large variance for μ is fine.
- Improper priors must be used cautiously for complicated models.
- A natural prior for the variances is

$$\tau^2, \sigma^2 \stackrel{IID}{\sim} \text{Inverse-Gamma}(a, b).$$

- All full conditional distribution are conjugate and MCMC sampling is very fast.

One-way random effects model

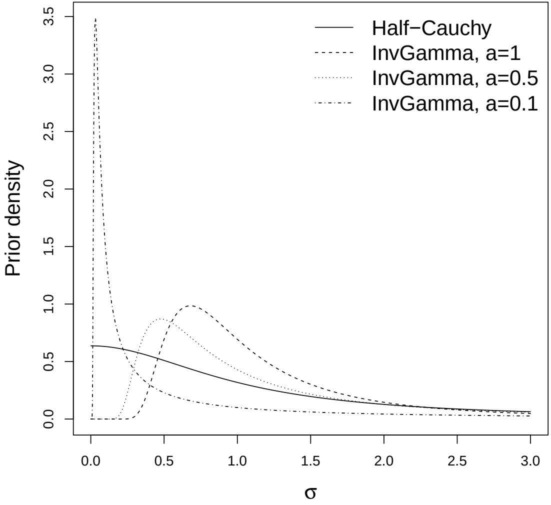
- However, under the inverse gamma prior for the variances the induced priors for σ and τ have no mass at zero.
- Gelman recommends the half-Cauchy prior for the SD

$$p(\sigma) = \frac{2}{\pi(1+\sigma^2)},$$

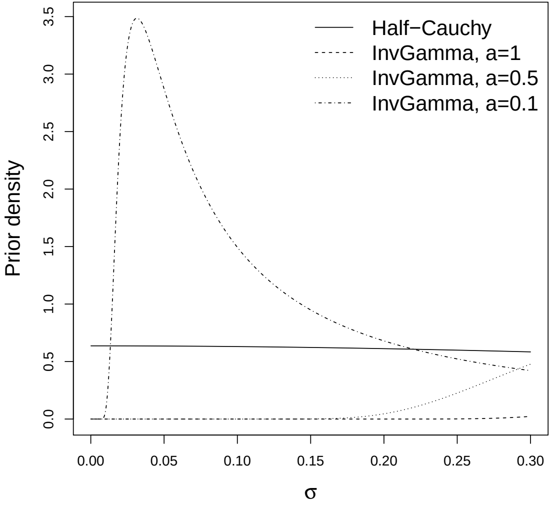
i.e., a Student-t density with 1 df restricted to be positive.

- This PDF is flat around zero and has heavy tails.
- This is very easy to code in JAGS.
- For large sample these give similar results, but half-Cauchy is often preferred.

Prior for standard deviation



Prior for standard deviation (zoomed in around 0)



Confusion about random effects

- MCMC does not distinguish between random effects and other parameters.
- For example, σ, τ, μ and α_j are all treated as random in a Bayesian analysis.
- However, α_j is called a "random" effect because it represents a random draw from the fixed Normal(μ, τ^2) population of classroom means.
- Try to implement Gibbs sampling for this model!

Linear mixed models

- Consider the model
$$Y_{ij} = \beta_0 + X_{ij}\beta_1 + \alpha_j + \varepsilon_{ij},$$
where X_{ij} is the age of student i in class j.

- The regression coefficients β_0 and β_1 apply to all students are all called “fixed effects”.

- The random effect is $\alpha_j \sim \text{Normal}(0, \tau^2)$.

- A linear model with both fixed and random effects is called a **linear mixed model**.

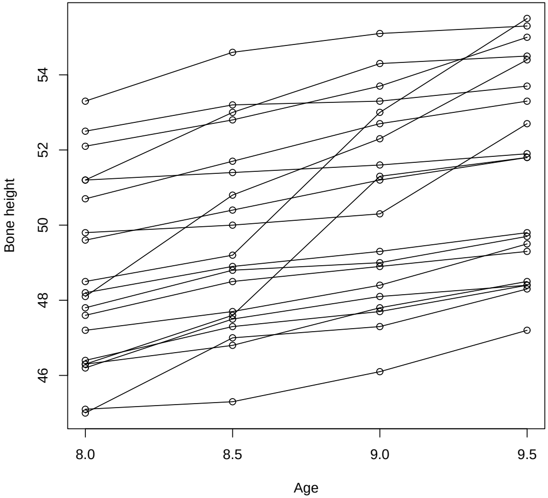
Random slopes model

- Let Y_{ij} be the j^{th} observation for subject i.
- As an example, consider the data plotted on the next slide were Y_{ij} is the bone height for child i at age X_j.
- Here we might specify a different regression for each child to capture variability over the population of children:

$$Y_{ij} \sim \text{Normal}(\gamma_{0i} + X_i\gamma_{1i}, \sigma^2).$$

- $\gamma_i = (\gamma_{i0}, \gamma_{i1})^T$ controls the growth curve for child i.
- These separate regression are tied together in the prior, $\gamma_i \sim \text{Normal}(\beta, \Sigma)$, which borrows strength across children.
- This is a linear mixed model: γ_i are random effects specific to one child and β are fixed effects common to all children.

Bone height data



Prior for a covariance matrix

- The random-effects covariance matrix is $\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix}$.
- σ_1^2 is the variance of the intercepts across children.
- σ_2^2 is the variance of the slopes across children.
- σ_{12} is the covariance between the intercepts and slopes.
- Prior 1: $\sigma_1^2, \sigma_2^2 \sim$ Inverse-Gamma and $\rho = \frac{\sigma_{12}}{\sigma_1 \sigma_2} \sim \text{Unif}(-1, 1)$.
- Prior 2: Inverse Wishart works better in higher dimensions.

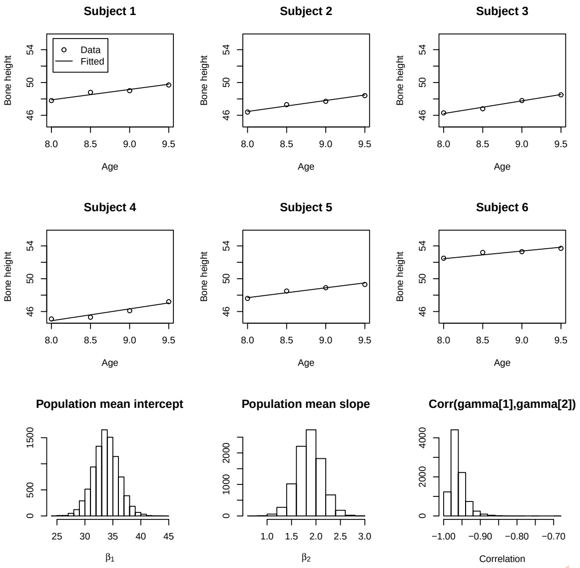
Inverse Wishart distribution

- The inverse Wishart distribution is the most common prior for a $p \times p$ covariance matrix.
- It reduces to the inverse gamma distribution if $p = 1$.
- Say $\Sigma \sim$ Inverse-Wishart(κ, R) where $\kappa > p + 1$ and R is a $p \times p$ covariance matrix. Both κ and R are hyperparameters.
- The PDF is
$$f(\Sigma) \propto |\Sigma|^{-(\kappa+p+1)/2} \exp\left[-\frac{1}{2}\text{trace}(R\Sigma^{-1})\right].$$
- The mean is $\frac{1}{\kappa-p-1}R$.

Full conditional distributions

- The hierarchical model is:
  - $Y_{ij} \sim \text{Normal}(\gamma_{0i} + X_i\gamma_{1i}, \sigma^2)$
  - $\gamma_i \sim \text{Normal}(\beta, \Sigma)$
  - $p(\beta) \propto 1$
  - $\sigma^2 \sim \text{Inverse-Gamma}(a, b)$
  - $\Sigma \sim \text{Inverse-Wishart}(\kappa, R)$
- The full conditionals are all in closed forms (check!).
- Try to write the JAGS code!

Bone height data - fitted values



Linear models with correlated errors

- An alternative to using random effects to capture dependence is to model correlation directly.
- For example, say the observations are collected at n different spatial locations.
- Denote the measurement at lat/lon s_i as Y_i.
- We might fit the model
$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i,$$
where the residual errors ε_i have spatial correlation.
- A common model is
$$\text{Cov}(\varepsilon_i, \varepsilon_j) = \sigma^2 \exp(-d_{ij}/\phi).$$
- The parameter ϕ controls the exponential decay of the correlation as distance between sites, d_{ij}, increases.

Linear models with correlated errors

- This is staightforward (though often slow) to fit using MCMC.
- The likelihood is multivariate normal
$$\mathbf{Y}|\beta, \sigma^2, \rho \sim \text{Normal}\left(\mathbf{X}\beta, \sigma^2\Sigma(\phi)\right).$$
- The $n \times n$ correlation matrix $\Sigma(\phi)$ has (i, j) element $\exp(-d_{ij}/\phi)$.
- This last piece is to set a prior for ϕ.
- A uniform prior between 0 and the maximum distance between points is an option.
- This type of modeling is also useful for time series data.

Thank you!

Lecture 22

Nonparametric regression

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Flexible regression modeling

- Nonparametric (NP) methods attempt to analyze the data by making the fewest number of assumptions as possible.
- NP methods are generally more robust and flexible, but less powerful than correctly specified parametric models.
- Most frequentist NP methods completely avoid specifying a model.
- For example, a rank or sign test to compare two means.

Non- and Semi-parametric modeling

- Bayesian methods need a likelihood in order to obtain a posterior, so you can't completely avoid specifying a model.
- Bayesian NP (BNP) then attempts to specify a model that is so flexible that it almost certainly captures the true model.
- One definition of the BNP model is one that has infinitely-many parameters.
- In some cases, NP models are difficult conceptually and computationally, and so semiparametric models with a large but finite number of parameters are useful approximations.

Parametric simple linear regression

Consider the classic parametric model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \quad \text{where} \quad \varepsilon_i \sim \mathrm{N}(0, \sigma^2).$$

Assumptions:

1. ε_i are independent
2. ε_i are Gaussian
3. The mean of Y_i is linear in X.
4. The residual distribution does not depend on X

Alternatives:

1. Parametric alternatives such as a time series model.
2. Let $\varepsilon_i \sim F$, and place a prior on the distribution F.
3. Let $E(Y|X) = g(X)$ and put a prior on the function g.
4. Heteroskedastic regression $\mathrm{Var}(\varepsilon_i) = \exp(\alpha_0 + \alpha_1 X)$.

In 2-4 we are placing priors on functions, not parameters.

Nonparametric regression

- Let's relax the assumption of linearity in the mean.

- The mean is $g(X)$, where g is some function that relates X to $\mathrm{E}(Y|X)$.

- Parametric models include
  1. Linear: $g(X) = \beta_0 + \beta_1 X$
  2. Quadratic: $g(X) = \beta_0 + \beta_1 X + \beta_2 X^2$
  3. Logistic: $g(X) = \beta_0 + \beta_1 \frac{\exp[\beta_2 + \beta_3 X]}{1 + \exp[\beta_2 + \beta_3 X]}$.

- NP regression puts a prior on the curve $g(X)$, rather than the parameters $\beta_1, ..., \beta_p$ that determine the parametric model.

Semiparametric regression

- Semiparametric regression approximates the function g using a finite basis expansion

$$g(X) = \sum_{j=1}^{J} B_j(X)\beta_j$$

  where $B_j(X)$ are known basis functions and β_j are unknown coefficients that determine the shape of g.
- Example: the cubic spline basis functions are

$$B_j(X) = (X - v_j)_+^3$$

  where v_j are fixed knots that span the range of X.
- Many other expansions exist: wavelets; Fourier, etc.
- Fact: A basis expansion of J terms can match the true curve g at any J points $X_1, ..., X_J$.
- So increasing J gives an arbitrarily flexible model.

Model fitting

- The model is $Y_i \sim \text{N}\left(B_i^T\boldsymbol{\beta}, \sigma^2\right)$, where $\beta_j \sim \text{N}(0, \tau^2)$ and B_i is comprised of the known basis functions $B_j(X_i)$.
- Therefore, the model is usual linear regression model and is straightforward to fit using MCMC.
- How to pick J?
- Can we have more basis functions than observations?

Thank you!

Lecture 23

Model comparison part 1

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Model selection

- So far we have many potential models in our toolbox.
- For a given dataset, how to determine whether a simple model is sufficient or if we need a very complex one?
- There is probably no “right” model in practice.
- A **statistical model** is just a mathematical representation of the system that includes errors and biases in the observational process.
- All models are basically some simplifications of the reality and the reality is never that simple.
- We want a model that is as simple as possible yet seems to explain the patterns in the data reasonably well.

Outline of Chapter 5

- For model comparisons, there are a finite number of candidate models and we want to select one using
  - Bayes factors
  - Stochastic search variable selection
  - Cross validation
  - Deviance information criteria (DIC)
  - Watanabe-Akaike information criteria (WAIC)
- In cases where multiple models fit well, we will discuss
  - Bayesian model averaging (BMA)
- After selecting a model, we want to test whether it fits the data well
  - Posterior predictive checks

Bayes factors (BF)

- In some sense, BFs are the gold standard.
- Suppose, we are comparing two models, $\mathcal{M}_1$ and $\mathcal{M}_2$.
- For example, $Y \sim \text{Binomial}(n, \theta)$ and the two models are

$$\mathcal{M}_1 : \theta = 0.5 \ \text{ and } \ \mathcal{M}_2 : \theta \neq 0.5.$$

- Another example would be $Y_1, Y_2, ..., Y_n$ being a time series and

$$\mathcal{M}_1 : \text{Cor}(Y_{t+1}, Y_t) = 0 \ \text{ and } \ \mathcal{M}_2 : \text{Cor}(Y_{t+1}, Y_t) > 0.$$

- Another example in the multiple linear regression setting would be

$$\mathcal{M}_1 : \text{E}(Y) = \beta_0 + \beta_1 X \ \text{ and } \ \mathcal{M}_2 : \text{E}(Y) = \beta_0 + \beta_1 X + \beta_2 X^2.$$

Bayes factors (BF)

- BF is basically the same as the frequentist hypothesis testing.
- We proceed by computing the posterior probability of the models $\mathcal{M}_1$ and $\mathcal{M}_2$.
- This require priors probabilities $p(\mathcal{M}_1)$ and $p(\mathcal{M}_2)$.
- Note that this is not prior on a parameter, rather it is a prior on the set of models!
- This approach permits conclusions like “Given the data we have observed, the quadratic model is 5 times more likely than a linear model”.

Bayes factors (BF)

- The BF for model $\mathcal{M}_2$ compared to model $\mathcal{M}_1$ is

$$BF_{2,1} = \frac{\text{Posterior odds}}{\text{Prior odds}} = \frac{p(\mathcal{M}_2|\mathbf{Y})/p(\mathcal{M}_1|\mathbf{Y})}{p(\mathcal{M}_2)/p(\mathcal{M}_1)} = \frac{p(\mathbf{Y}|\mathcal{M}_2)}{p(\mathbf{Y}|\mathcal{M}_1)}.$$

- Rule of thumb: $BF_{2,1} > 10$ is strong evidence for $\mathcal{M}_2$.

- Rule of thumb: $BF_{2,1} > 100$ is decisive evidence for $\mathcal{M}_2$.

- In linear regression, *BIC* approximates the log-BF comparing a model to the null model.

Example

- $Y \sim \text{Binomial}(N, \theta)$ with
$$\mathcal{M}_1 : \theta = 0.5 \ \text{ and } \ \mathcal{M}_2 : \theta \neq 0.5.$$
- $p(Y|\mathcal{M}_1)$ is just the binomial density with $\theta = 0.5$.
- $\mathcal{M}_2$ involves an unknown parameter θ.
- This requires a prior, say $\theta \sim \text{Beta}(a, b)$, and integration
$$p(Y|\mathcal{M}_2) = \int p(Y, \theta) d\theta = \int p(Y|\theta) p(\theta) d\theta.$$

Calculation of $p(Y|\mathcal{M}_2)$

- Under $\mathcal{M}_2$, we have

  Likelihood: $Y|\theta \sim \text{Binomial}(N, \theta)$,
  Prior: $\theta \sim \text{Beta}(a, b)$.

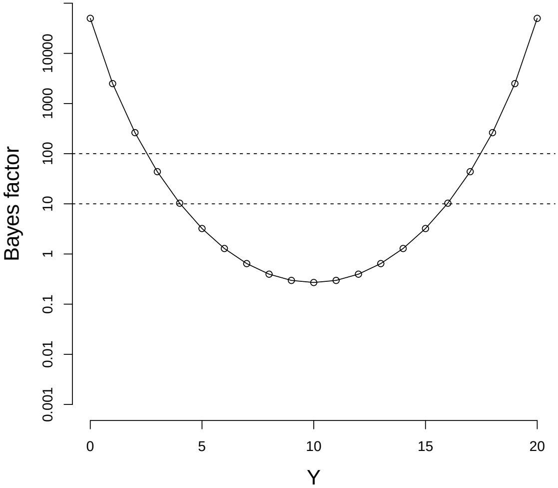
- Thus,

$$\begin{aligned} p(Y|\mathcal{M}_2) &= \int p(Y, \theta) d\theta = \int p(Y|\theta) p(\theta) d\theta \\ &= \int \binom{N}{Y} \theta^Y (1-\theta)^{N-Y} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1-\theta)^{b-1} d\theta \\ &= \binom{N}{Y} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \int \theta^{Y+a-1} (1-\theta)^{N-Y+b-1} d\theta \\ &= \binom{N}{Y} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \frac{\Gamma(Y+a)\Gamma(N-Y+b)}{\Gamma(N+a+b)}. \end{aligned}$$

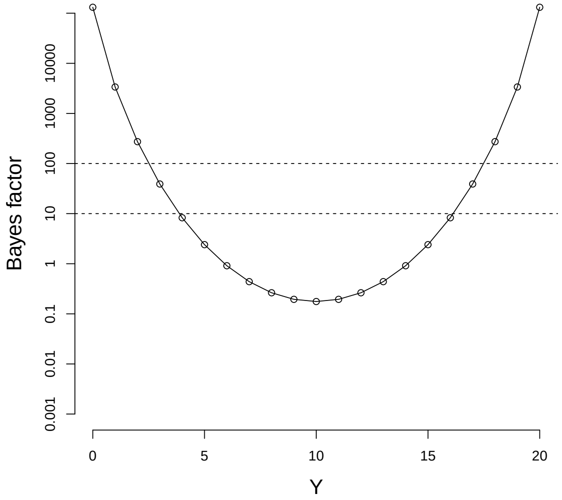
Calculation of $BF_{2,1}$

- $$\begin{aligned} BF_{2,1} &= \frac{p(Y|\mathcal{M}_2)}{p(Y|\mathcal{M}_1)} \\ &= \frac{\binom{N}{Y}\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\frac{\Gamma(Y+a)\Gamma(N-Y+b)}{\Gamma(N+a+b)}}{\binom{N}{Y}0.5^Y 0.5^{N-Y}} \\ &= \frac{\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\frac{\Gamma(Y+a)\Gamma(N-Y+b)}{\Gamma(N+a+b)}}{0.5^Y 0.5^{N-Y}}. \end{aligned}$$

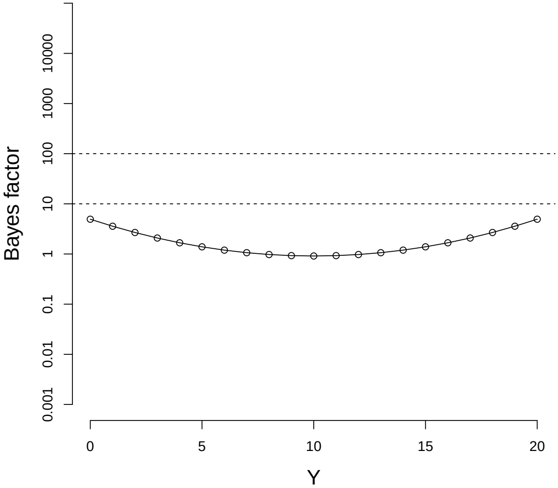
BF by Y with $N = 20$ and prior $a = 1$ and $b = 1$



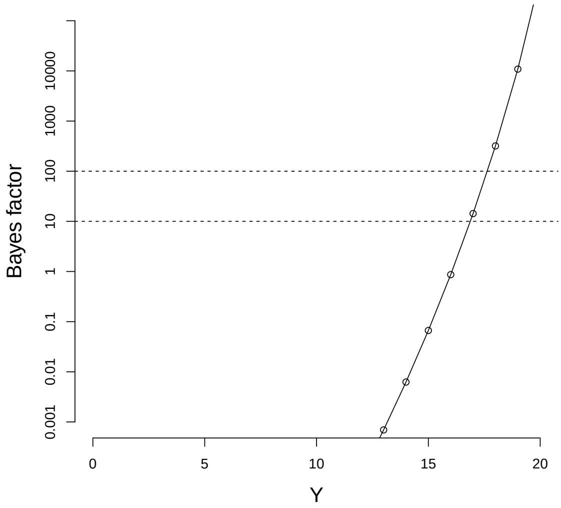
BF by Y with $N = 20$ and prior $a = 0.5$ and $b = 0.5$



BF by Y with $N = 20$ and prior $a = 50$ and $b = 50$



BF by Y with $N = 20$ and prior $a = 50$ and $b = 1$



Problems with Bayes factors

- Often hard to compute the required integrals which is only feasible for simple models.
- Requires proper priors.
- Can be very sensitive to priors (Lindley's paradox).
- In most cases, computing posterior credible intervals from the full model and then testing by comparing these to the null is preferred.

Computing Bayes factors using MCMC

- If models can be written as nested, then MCMC can be used to approximate model probabilities.
- For example, say

$$\mathcal{M}_1 : \mathrm{E}(Y) = \beta_0 + \beta_1 X \text{ and } \mathcal{M}_2 : \mathrm{E}(Y) = \beta_0 + \beta_1 X + \beta_2 X^2.$$

- Both model can be written as

$$\mathrm{E}(Y) = \beta_0 + \beta_1 X + \gamma\beta_2 X^2,$$

  where $\gamma \in \{0, 1\}$ indicates the model.
- The prior on models becomes $\gamma \sim \text{Bernoulli}(0.5)$.
- Then $\text{Prob}(\gamma = 1|\mathbf{Y}) = \text{Prob}(\mathcal{M}_2|\mathbf{Y})$ can be appoximated using MCMC.

Stochastic search variable selection (SSVS)

- This is the Bayesian analog of forward/backward/stepwise variable selection.
- We place a prior on all 2^p models using p variable inclusion indicators γ_j.
- MCMC returns the approximate posterior probability of each model.
- With large p all models will have low probability and so this requires long MCMC runs.
- Similar to Bayes factor, SSVS can be sensitive to priors.

Model averaging

- Let us go back to the linear regression example

$$\mathcal{M}_1 : \mathrm{E}(Y) = \beta_0 + \beta_1 X \text{ and } \mathcal{M}_2 : \mathrm{E}(Y) = \beta_0 + \beta_1 X + \beta_2 X^2.$$

- Say we have fit both models and found that both are about equally likely, but that $\mathcal{M}_1$ is slightly preferred.
- For prediction, $\hat{Y}$, we could simply take the prediction that comes from fitting $\mathcal{M}_1$.
- But the prediction from $\mathcal{M}_2$ is likely different and nearly as accurate.
- Also, taking the prediction from $\mathcal{M}_1$ suppresses our uncertainty about the form of the model.

Model averaging

- Let $\hat{Y}_k$ be the prediction from model $\mathcal{M}_k$ for $k = 1, 2$.
- The model averaged predictor is
$$\hat{Y} = w\hat{Y}_1 + (1 - w)\hat{Y}_2.$$
- It can be shown that the optimal weight w is the posterior probability of $\mathcal{M}_1$.
- Averaging adds stability.
- In linear regression with p predictors the prediction is a weighted average of 2^p possible models.
- This is implemented in the `R` package `BMA`.

Thank you!

Lecture 24

Model comparison part 2

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Recap: Model selection

- So far we have many potential models in our toolbox.
- For a given dataset, how to determine whether a simple model is sufficient or if we need a very complex one?
- There is probably no “right” model in practice.
- A **statistical model** is just a mathematical representation of the system that includes errors and biases in the observational process.
- All models are basically some simplifications of the reality and the reality is never that simple.
- We want a model that is as simple as possible yet seems to explain the patterns in the data reasonably well.

Outline of Chapter 5

- For model comparisons, there are a finite number of candidate models and we want to select one using
  - Bayes factors
  - Stochastic search variable selection
  - Cross validation
  - Deviance information criteria (DIC)
  - Watanabe-Akaike information criteria (WAIC)
- In cases where multiple models fit well, we will discuss
  - Bayesian model averaging (BMA)
- After selecting a model, we want to test whether it fits the data well
  - Posterior predictive checks

Recap: Bayes factors (BF)

- The BF for model $\mathcal{M}_2$ compared to model $\mathcal{M}_1$ is

$$BF_{2,1} = \frac{\text{Posterior odds}}{\text{Prior odds}} = \frac{p(\mathcal{M}_2|\mathbf{Y})/p(\mathcal{M}_1|\mathbf{Y})}{p(\mathcal{M}_2)/p(\mathcal{M}_1)} = \frac{p(\mathbf{Y}|\mathcal{M}_2)}{p(\mathbf{Y}|\mathcal{M}_1)}.$$

- Rule of thumb: $BF_{2,1} > 10$ is strong evidence for $\mathcal{M}_2$.
- Rule of thumb: $BF_{2,1} > 100$ is decisive evidence for $\mathcal{M}_2$.

Cross validation

- A very common approach for model selection is cross-validation.
- This is exactly the same procedure used in classical statistics.
- This operates under the assumption that the “true” model likely produces better out-of-sample predictions than competing models.
- Advantages: Simple, intuitive, and broadly applicable.
- Disadvantages: Slow because it requires several model fits and it is hard to say a difference is statistically significant.

K-fold Cross validation

0. Split the data into K equally-sized groups
1. Set aside group k as test set and fit the model to the remaining $K - 1$ groups
2. Make predictions for the test set k based on the model fit to the training data
3. Repeat steps 1 and 2 for $k = 1, ..., K$ giving a predicted value $\hat{Y}_i$ for all n observations
4. Measure prediction accuracy, e.g.,

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

Variants

- Usually K is either 5 or 10.
- $K = n$ is called "leave-one-out" cross-validation, which is great but slow.
- The predicted value $\hat{Y}_i$ can be either the posterior predictive mean or median.
- Mean squared error (MSE) can be replaced with Mean absolute deviation

$$MAD = \frac{1}{n}\sum_{i=1}^{n}|Y_i - \hat{Y}_i|.$$

Information criteria

- Several information criteria have been proposed that do not require fitting the model several times.
- Many are functions of the **deviance**, i.e., twice the negative log likelihood

$$D(\mathbf{Y}|\theta) = -2\log[f(\mathbf{Y}|\theta)].$$

- Ideally, models will have small deviance.
- However, if a model is too complex it will have small deviance (overfitting).
- The Akaike information criteria has a complexity penalty

$$AIC = D(\mathbf{Y}|\hat{\theta}) + 2p,$$

  where $\hat{\theta}$ is the MLE.
- Model with smaller AIC are preferred.

Bayesian information criteria (BIC)

- The Bayesian information criteria is similar

$$BIC = D(\mathbf{Y}|\hat{\theta}) + \log(n)p.$$

- This is motivated as an approximation to the log Bayes factor of the model compared to the null model.
- However, this is only an asymptotic (large n) approximation.
- With large n the prior is irrelevant, and so this is not satisfying to a subjective Bayesian.

Deviance information criteria (DIC)

- *DIC* is a popular Bayesian analog of *AIC* or *BIC*.
- Unlike CV, *DIC* requires only one model fit.
- Unlike BF, it can be applied to complex models.
- However, proceed with caution.
- *DIC* really only applies when the posterior is approximately normal, and will give misleading results when the posterior far from normality, e.g., bimodal.
- *DIC* is also criticized for selecting overly-complex models.

Deviance information criteria (DIC)

- Let $\bar{D} = \mathrm{E}[D(Y|\theta)|\mathbf{Y}]$ be the posterior mean of the deviance.
- Denote $\hat{\boldsymbol{\theta}}$ as the posterior mean of $\boldsymbol{\theta}$.
- The effective number of parameters is

$$p_D = \bar{D} - D(\mathbf{Y}|\hat{\theta}).$$

- DIC can be written like AIC,

$$DIC = \bar{D} + p_D = D(\mathbf{Y}|\hat{\theta}) + 2p_D.$$

- Models with small $\bar{D}$ fit the data well.
- Models with small p_D are simple.
- We prefer models that are simple and fit well, so we select the model with smallest DIC.

DIC

- The effective number of parameters is a useful measure of model complexity.
- Intuitively, if there are p parameters and we have uninformative priors then $p_D \approx p$.
- However, $p_D << p$ if there are strong priors.
- For example, how many free degrees of freedom do we have with $\theta \sim \text{Beta}(1, 1)$ versus $\theta \sim \text{Beta}(1000, 1000)$?
- In some cases p_D has a nice closed form.

DIC

- As with *AIC* or *BIC*, we compute *DIC* for all models under consideration and select the one with smallest *DIC*.
- Rule of thumb: a difference of *DIC* of less than 5 is not definitive and a difference greater than 10 is substantial.
- As with *AIC* or *BIC*, the actual value is meaningless, only differences are relevant.
- *DIC* can only be used to compare models with the same likelihood.

Watanabe-Akaike information criteria (WAIC)

- *WAIC* is an alternative to *DIC*.
- It is motivated as an approximation to leave-one-out CV.
- In the end *WAIC* has model-fit and model-complexity components.
- It is used the same as *DIC* with smaller *WAIC* begin preferred.
- In practice the two often give similar results, but *WAIC* is arguably more theoretically justified.

Watanabe-Akaike information criteria (WAIC)

- *WAIC* is written in terms of the posterior of the likelihood rather than parameters.
- Let m_i and v_i be the posterior mean and variance of

$$\log[f(Y_i|\theta)].$$

- The effective model size is $p_W = \sum_{i=1}^n v_i$.
- The criteria is

$$WAIC = -2\sum_{i=1}^n m_i + 2p_W.$$

Thank you!

Lecture 25

Posterior predictive checks

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Posterior predictive checks

- After comparing a few models, we settle on the one that seems to fit the best.
- Given this best model, we then verify if it is adequate.
- The usual residual checks are appropriate here: e.g., qq-plots.
- A uniquely Bayesian diagnostic is the posterior predictive check.
- This leads to the Bayesian p-value.

Posterior predictive distributions

- Before discussing posterior predictive checks, let us review Bayesian prediction in general.

- The plug-in approach would fix the parameters θ at the posterior mean $\hat{\theta}$ and then predict $Y_{new} \sim f(y|\hat{\theta})$.

- This suppresses uncertainty in θ.

- We would like to propagate this uncertainty through to the predictions.

Posterior predictive distributions (PPD)

- We really want the PPD

$$f(Y_{new}|\mathbf{Y}) = \int f(Y_{new}, \theta|\mathbf{Y})d\theta = \int f(Y_{new}|\theta)f(\theta|\mathbf{Y})d\theta.$$

- MCMC easily produces draws from this distribution.

- To make S draws from the PPD, for each of the S MCMC samples of θ, we draw a Y_{new}.

- This gives draws from the PPD and clearly accounts for uncertainty in θ.

Posterior predictive checks

- Posterior predictive checks sample many datasets from the PPD with the identical design (same n, same $\mathbf{X}$) as the original data set.
- We then define a statistic describing the dataset, e.g.,

$$d(\mathbf{Y}) = \max\{Y_1, ..., Y_n\}.$$

- Denote the statistic for the original data set as d_0 and the statistic from simulated data set number s as d_s.
- If the model is correct, then d_0 should fall in the middle of the $d_1, ..., d_S$.

Posterior predictive checks

- A measure of how extreme the observed data is relative to this sampling distribution is the Bayesian *p*-value.

$$p = \frac{1}{S}\sum_{s=1}^{S}\mathbb{I}(d_s > d_0).$$

- If p is near zero or one the model doesn't fit.

- This is repeated for several d to give a comprehensive evaluation of model fit.

Thank you!

Lecture 26

Bayesian Hierarchical models

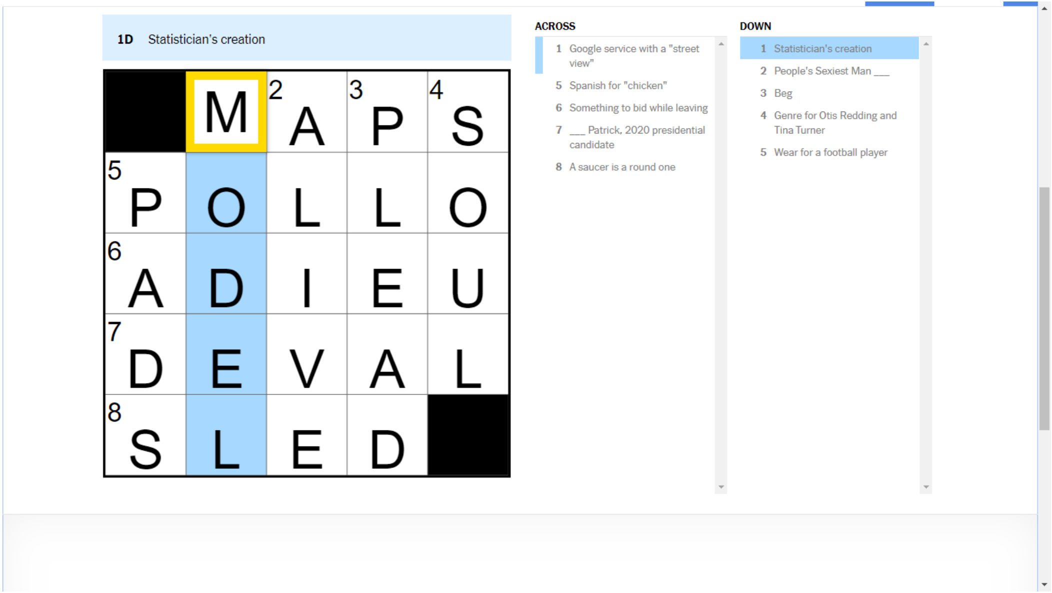
Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Hierarchical models

- Hierarchical modeling provides a framework for building complex and high-dimensional models from simple and low-dimensional building blocks.
- Of course, it is possible to analyze these models using non-Bayesian methods.
- However, this modeling framework is popular in the Bayesian literature because MCMC is conducive to hierarchical models.
- Both “divide and conquer” big problems by splitting them into a series of smaller problems in the same way.

We build models!



Outline oc Chapter 6

- Building a hierarchical model through layers
- Directed acyclic graphs
- Several examples

Hierarchical models

Often Bayesian models can we written in the following layers of the hierarchy

1. **Data layer**: $[\mathbf{Y}|\theta, \alpha]$ is the likelihood for the observed data **Y** given the model parameters.

2. **Process layer**: $[\theta|\alpha]$ is the model for the parameters $\boldsymbol{\theta}$ that define the latent data generating process.

3. **Prior layer**: $[\alpha]$ prior for hyperparameters.

Epidemiology example - Data layer

- Let S_t and I_t be the number of susceptible and infected individuals in a population, respectively, at time t.
- The data Y_t is the number of observed cases at time t.
- The data layer models our ability to measure the process I_t.
- **Data layer**: $Y_t | I_t \sim \text{Binomial}(I_t, p)$.
- This assumes no false positives and false negative probability p.

Epidemiology example - Process layer

- Scientific understanding of the disease is used to model disease propagation.
- We might select the simple Reed-Frost model

$$\textbf{Process layer:} \qquad I_{t+1} \sim \text{Binomial}\left[S_t, 1-(1-q)^{I_t}\right], \\ S_{t+1} = S_t - I_{t+1}.$$

- This assumes all infected individuals are removed from the population before the next time step.
- Also that q is the probability of a non-infected person coming into contact with and contracting the disease from an infected individual.

Epidemiology example - Prior layer

- The epidemiological process-layer model expresses the disease dynamics up to a few unknown parameters.
- The Bayesian model is completed using priors, say,
- **Prior layer:**

$$\begin{aligned} I_1 &\sim \text{Poisson}(\lambda_1), \\ S_1 &\sim \text{Poisson}(\lambda_2), \\ p, q &\sim \text{beta}(a, b). \end{aligned}$$

When to stop adding layers?

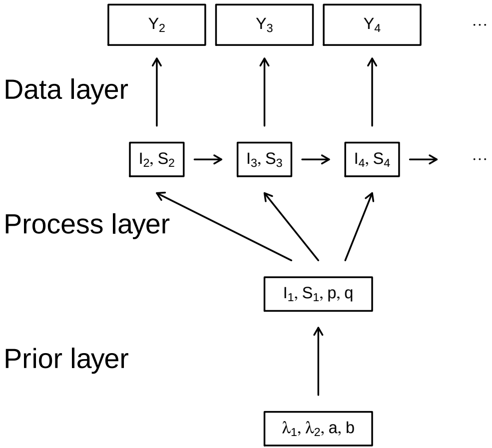
- In the previous example a, b, λ_1 and λ_2 are fixed.
- But we will have uncertainty about the correct value.
- Maybe replace a fixed value with another layer, say $a \sim \text{Uniform}(0, \theta)$?
- Then maybe $\theta \sim \text{Exponential}(\xi)$, $\xi \sim \text{Uniform}(0, \eta)$, etc.
- Rule of thumb: Be careful assigning priors to parameters in layers without replication.
- For example, even if we knew p exactly this would be just one value and we couldn't hope to estimate the parameters of its beta distribution.

Directed acyclic graphs (DAGs)

- A DAG is a graphical representation of a hierarchical model.
- DAGS sometimes go by the name Bayesian networks.
- Each observation and parameter is a node.
- An arrow for X to Y means that the conditional distribution of Y depends on X.
- "Directed" means that arrows only go one way.
- Acyclic means there are no cycles, e.g.,

$$X \to Y \to Z \to X.$$

Epidemiology example - DAG



Directed acyclic graphs (DAGs)

- Building models this way ensures we will always have a valid joint distribution.
- For example, say we need to specify the joint distribution of (X, Y, Z).
- Any joint distribution can be written as

$$f(X, Y, Z) = f(X)f(Y|X)f(Z|X, Y).$$

- This is a fully-connected DAG.
- Ad-hoc constructions like

$$f(X, Y, Z) = f(X|Z)f(Y|X)f(Z|X, Y)$$

may or may not give a valid joint PDF.

Hierarchical models and MCMC

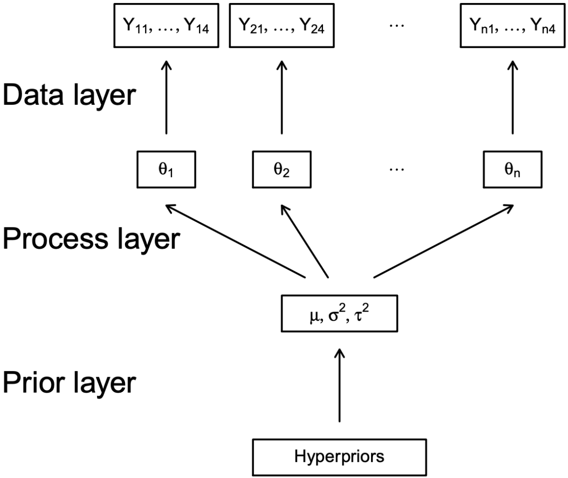
- Consider the classic one-way random effects model:

$$Y_{ij} \sim \mathrm{N}(\theta_i, \sigma^2) \quad \text{and} \quad \theta_i \sim N(\mu, \tau^2)$$

  where Y_{ij} is the j^{th} replicate for unit i and $\boldsymbol{\alpha} = (\mu, \sigma^2, \tau^2)$ has an uninformative prior.

- This hierarchy can be written using a directed acyclic graph.

Random effects example - DAG



Hierarchical models and MCMC

- MCMC is efficient in this case even if the number of parameter or levels of the hierarchy is large.
- You only need to consider "connected nodes" when you update each parameter.
- 1. $[\theta_i|\cdot]$
  2. $[\mu|\cdot]$
  3. $[\sigma^2|\cdot]$
  4. $[\tau^2|\cdot]$
- Each of these updates is a draw from a standard one-dimensional normal or inverse gamma.

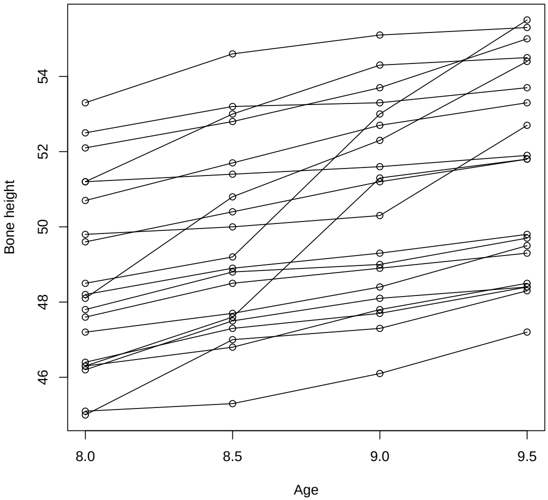
Random slopes model

- Let Y_{ij} be the j^{th} observation for subject i.
- As an example, consider the data plotted on the next slide were Y_{ij} is the bone density for child i at age X_j.
- Here we might specify a different regression for each child to capture variability over the population of children:

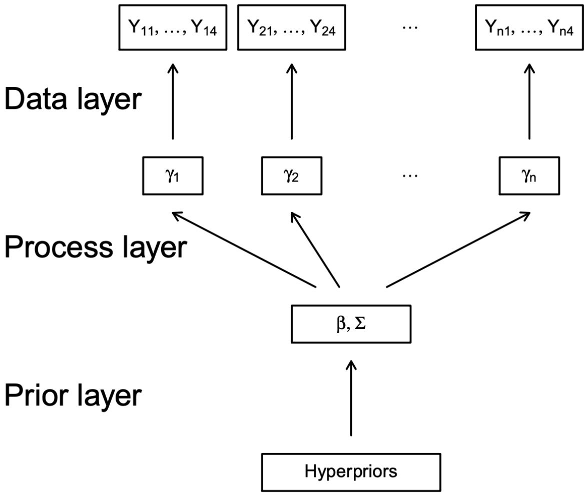
$$Y_{ij} \sim \text{Normal}(\gamma_{0i} + X_j\gamma_{1i}, \sigma^2).$$

- $\gamma_i = (\gamma_{i0}, \gamma_{i1})^T$ controls the growth curve for child i.
- These separate regression are tied together in the prior, $\gamma_i \sim \text{Normal}(\beta, \Sigma)$, which borrows strength across children.
- This is a linear mixed model: γ_i are random effects specific to one child and β are fixed effects common to all children.

Bone height data



Draw a DAG and work out the full conditionals



Thank you!

Lecture 27

Bayesian missing data imputation

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Recap: Hierarchical models

Often Bayesian models can be written in the following layers of the hierarchy

1. **Data layer**: $[\mathbf{Y}|\theta, \alpha]$ is the likelihood for the observed data $\mathbf{Y}$ given the model parameters.

2. **Process layer**: $[\theta|\alpha]$ is the model for the parameters $\boldsymbol{\theta}$ that define the latent data generating process.

3. **Prior layer**: $[\alpha]$ prior for hyperparameters.

Missing data models

- We will deal with missing data in the linear regression context, but the ideas apply to all models.
- The model is
$$Y_i \sim \text{Normal}(\beta_0 + \beta_1 X_{i1} + ... + \beta_p X_{ip}, \sigma^2).$$
- Often either Y_i or elements X_{ij} are missing.
- We will study separately the case of missing responses and missing covariates.

Missing responses

- If the response is missing this is essentially a prediction problem.
- We obtain samples from the PPD of Y_i.
- At each MCMC iteration we simply draw

$$Y_i \sim \text{Normal}(\beta_0 + \beta_1 X_{i1} + ... + \beta_p X_{ip}, \sigma^2).$$

- This distribution accounts for random error as well as uncertainty in the model parameters.
- For the other updates the data are essentially complete.

Missing covariates

- Now say all responses are observed, but a some covariates are missing.
- The simplest approach is imputation, e.g., just plug in the sample mean of the covariate for the missing values.
- This doesn't account for uncertainty in the imputations.
- Bayesian methods handle this well using MCMC.

Missing covariates

- The main idea is to treat the missing values as unknown parameters in the Bayesian model.
- Unknown parameters need priors, so missing $\mathbf{X}_i = (X_{i1}, ..., X_{ip})^T$ must have priors such as
$$\mathbf{X}_i \sim \text{Normal}(\boldsymbol{\mu}_X, \boldsymbol{\Sigma}_X).$$
- Assumptions about missing data:
  - Missing status is independent of Y_i and $\mathbf{X}_i$
  - Covariates are Gaussian
- There are ways to relax both assumptions, but it becomes complicated.

Missing covariates

- Of course if the prior is way off, the results will be invalid.
- For example, if in reality the data are not missing at random the Bayesian model will likely give bad results.
- If specified correctly, the model will lead to inference for β that properly accounts for uncertainty about the missing data.

Hierarchical linear regression model with missing data

- $Y_i | \mathbf{X}_i, \boldsymbol{\beta}, \sigma^2 \sim \text{Normal}(\mathbf{X}_i^T \boldsymbol{\beta}, \sigma^2)$
- $\mathbf{X}_i | \boldsymbol{\mu}, \boldsymbol{\Sigma} \sim \text{Normal}_P(\boldsymbol{\mu}, \boldsymbol{\Sigma})$
- $p(\boldsymbol{\beta}) \propto \text{Normal}(0, 100^2 \boldsymbol{I}_P)$
- $\sigma^2 \sim \text{Inverse-Gamma}(0.01, 0.01)$
- $\boldsymbol{\mu} \sim \text{Normal}(0, 100^2 \boldsymbol{I}_P)$
- $\boldsymbol{\Sigma} \sim \text{Inverse-Wishart}(0.01, 0.01 \boldsymbol{I}_P)$

Overview of the Gibbs sampling algorithm

- The full conditional of missing Y_i is:

$$Y_i|\mathbf{X}_i, \boldsymbol{\beta}, \sigma^2 \sim \text{Normal}(\mathbf{X}_i^T\boldsymbol{\beta}, \sigma^2)$$

- The full conditional of missing X_i is:

  conjugate and similar to that of $\boldsymbol{\beta}$, and the derivation follows the same path as that for $\boldsymbol{\beta}$.

- In fact, all the full conditionals are conjugate.

Thank you!

Lecture 28

Frequentist properties of Bayesian methods

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Calibrated Bayes

- So far we have discussed Bayesian methods as being separate from the frequentist approach.
- However, in many cases, methods with frequentist properties are desirable.
- For example, we may want a method with Type-I error control or 80% power.
- You can design Bayesian methods to achieve these frequentist properties.
- In this view, Bayesian methods generate procedures/algorithms for further study.
- Often Bayesian methods are very competitive with frequentist methods using frequentist criteria.

Outline of Chapter 7

- Decision theory
- Bias-variance tradeoff
- Asymptotics
- Simulation studies

Should Bayesians care about frequentist properties?

What if a Bayesian weather forecaster made a 95% prediction interval for temperature every day for a year but the interval only included the actual temperature 40% of the time?

Little, 2011, Stat Sci

Statistical Science
2011, Vol. 26, No. 2, 162–174
DOI: 10.1214/10-STS318

Calibrated Bayes, for Statistics in General, and Missing Data in Particular[1]

Roderick Little

Abstract. It is argued that the Calibrated Bayesian (CB) approach to statistical inference capitalizes on the strength of Bayesian and frequentist approaches to statistical inference. In the CB approach, inferences under a particular model are Bayesian, but frequentist methods are useful for model development and model checking. In this article the CB approach is outlined. Bayesian methods for missing data are then reviewed from a CB perspective. The basic theory of the Bayesian approach, and the closely related technique of multiple imputation, is described. Then applications of the Bayesian approach to normal models are described, both for monotone and nonmonotone missing data patterns. Sequential Regression Multivariate Imputation and Penalized Spline of Propensity Models are presented as two useful approaches for relaxing distributional assumptions.

Key words and phrases: Maximum likelihood, multiple imputation, penalized splines, propensity models, sequential regression multivariate imputation.

Little in *Little, 2011, Stat Sci*

- Bayesian statistics is strong for inference under an assumed model, but relatively weak for the development and assessment of models.
- Frequentist statistics provides useful tools for model development and assessment, but has weaknesses for inference under an assumed model.
- If this summary is accepted, then the natural compromise is to use frequentist methods for model development and assessment, and Bayesian methods for inference under a model.
- This capitalizes on the strengths of both paradigms, and is the essence of the approach known as Calibrated Bayes.

Rubin on *Little, 2011, Stat Sci*

- The applied statistician should be Bayesian in principle and calibrated to the real world in practice - appropriate frequency calculations help to define such a tie.

- Frequency calculations are useful for making Bayesian statements scientific, scientific in the sense of capable of being shown wrong by empirical test.

- Here the technique is the calibration of Bayesian probabilities to the frequencies of actual events.

Bayes as a procedure generator

- A Bayesian analysis produces a posterior distribution which summarize our uncertainty after observing the data.
- However, if you have to give a one-number summary as an estimate you might pick the posterior mean

$$\hat{\theta}_B = \mathsf{E}(\theta|\mathbf{Y}).$$

- This estimator $\hat{\theta}_B$ can be evaluated along with MLE or method of moments estimators.
- Is it biased? Consistent? How does its MSE compare with the MLE?
- These are all frequentist properties of the Bayesian estimator.

Bayes as a procedure generator

- Similarly, if we have to give an interval estimate, we might use the 95% posterior credible set.
- In practice, this interval is motivated by the one data set we observed.
- But we could view this as a procedure for constructing an interval and inspect its frequentist properties.
- If we analyzed many datasets, each time computing a 95% posterior interval, how many would contain the true value?
- A Bayes test is to reject H_o if Prob$(H_o|\mathbf{Y}) < c$.
- What are the Type I and Type II errors of this test?
- Can we pick the threshold c to control Type I error?

Bayesian decision theory

- Before studying the frequentist properties of Bayesian estimtors and hypothesis tests, we should determine the "best" Bayesian method.
- For example, should we take the estimator to be the posterior mean, median, or mode?
- Defining "best" requires a scoring system.
- We call this the loss function $l(\hat{\theta}, \theta)$.
- Squared error loss is $l(\hat{\theta}, \theta) = (\hat{\theta} - \theta)^2$.
- Absolute loss is $l(\hat{\theta}, \theta) = |\hat{\theta} - \theta|$.

Bayesian decision theory

- The summary of the posterior that minimizes the expected (posterior) loss is the **Bayes rule.**
- Squared error loss implies we should use the posterior mean for $\hat{\theta}$.
- Absolute loss implies we should use the posterior median for $\hat{\theta}$.
- Hypothesis test requires are more complicated loss function.

Squared error loss

Let $\bar{\theta} = E_{\theta|Y}(\theta)$ denotes the posterior mean.

$$
\begin{aligned}
E_{\theta|Y}[l(\hat{\theta},\theta)] &= E_{\theta|Y}[(\hat{\theta}-\theta)^2] \\
&= E_{\theta|Y}[(\hat{\theta}-\bar{\theta}+\bar{\theta}-\theta)^2] \\
&= E_{\theta|Y}[(\hat{\theta}-\bar{\theta})^2] + 2E_{\theta|Y}[(\hat{\theta}-\bar{\theta})(\bar{\theta}-\theta)] + E_{\theta|Y}[(\bar{\theta}-\theta)^2] \\
&= (\hat{\theta}-\bar{\theta})^2 + 2(\hat{\theta}-\bar{\theta})E_{\theta|Y}[(\bar{\theta}-\theta)] + E_{\theta|Y}[(\bar{\theta}-\theta)^2] \\
&= (\hat{\theta}-\bar{\theta})^2 + V_{\theta|Y}[\theta]
\end{aligned}
$$

Here $V_{\theta|Y}[\theta]$ does not depend on θ. Thus, $E_{\theta|Y}[l(\hat{\theta},\theta)]$ is minimized when $\hat{\theta} = \bar{\theta}$.

Absolute error loss

We want to minimize $E_{\theta|Y}(|\hat{\theta}-\theta|) = \int |\hat{\theta}-\theta|\pi(\theta|Y)d\theta$.

$$
\begin{aligned}
E_{\theta|Y}(|\hat{\theta}-\theta|) &= \int |\hat{\theta}-\theta|\pi(\theta|Y)d\theta \\
&= \int_{-\infty}^{\hat{\theta}} |\hat{\theta}-\theta|\pi(\theta|Y)d\theta + \int_{\hat{\theta}}^{\infty} |\hat{\theta}-\theta|\pi(\theta|Y)d\theta \\
&= \int_{-\infty}^{\hat{\theta}} (\hat{\theta}-\theta)\pi(\theta|Y)d\theta + \int_{\hat{\theta}}^{\infty} (\theta-\hat{\theta})\pi(\theta|Y)d\theta
\end{aligned}
$$

Differentiate both the sides with respect to $\hat{\theta}$ and equate to zero. We get

$$
\int_{-\infty}^{\hat{\theta}} \pi(\theta|Y)d\theta = \int_{\hat{\theta}}^{\infty} \pi(\theta|Y)d\theta
$$

Thus, $\hat{\theta}$ is the median of $\pi(\theta|Y)$.

Bias/variance trade-off

- Assume $Y_1, ..., Y_n \sim \text{Normal}(\mu, \sigma^2)$.
- Estimator 1: $\hat{\mu}_1 = \bar{Y}$.
- Estimator 2: $\hat{\mu}_2 = c\bar{Y}$ where $c = \frac{n}{n+m}$.
- $\hat{\mu}_2$ is the posterior mean under prior $\mu \sim \text{Normal}(0, \frac{\sigma^2}{m})$.
- Compute the bias and variance of each estimator.
- Compute the mean squared error (recall MSE = bias^2+variance).
- Which estimator is preferred?

Properties of Bayesian estimators

Broadly speaking, the following comparisons between Bayes and MLE hold:

- Bayesian estimators have smaller standard errors because the prior adds information.
- Bayesian estimators are biased if the prior is not centered on the truth.
- Depending on this bias/variance trade-off, Bayes estimators may have smaller MSE than the MLE.
- If the prior is weak the methods are similar.
- For any prior that does not depend on the sample size, as n increases the prior is overwhelmed by the likelihood and the posterior approaches the MLE's sampling distribution.

Bayesian central limit theorem

- Assumptions:
  - the usual MLE conditions on the likelihood.
  - the prior does not depend on n and puts non-zero probability on the true value θ_0.
- Then
$$p(\theta|\mathbf{Y}) \rightarrow \text{Normal}\left[\theta_0, I(\theta_0)^{-1}\right],$$
where I is the information matrix.
- Therefore, for large datasets the posterior is approximately normal.
- Bayes methods are asymptotically unbiased.

Bayesian central limit theorem

- This implies that Bayes and MLE will be equivalent in large samples.
- What a relief!
- However, the interpretation is different.
- We can use the Bayesian interpretation like Prob($\mathcal{H}_0|\mathbf{Y}$) and Prob($3.4 < \theta < 5.6$).
- The Bayesian CLT gives a way to approximate ($n \to \infty$) the posterior without MCMC.
- Most still use MCMC with the hope that it better approximates ($S \to \infty$) the exact posterior.
- The CLT is useful for initial values and tuning.

Methods for studying frequentist properties

- Theoretical studies of Bayesian estimators use the same basic approaches as frequentist methods.
- Theorems and proofs (of consistency etc.) are ideal.
- When the math is intractable, simulation studies are used.
- In a simulation study you generate many datasets with known parameters values.
- You apply the Bayesian method to each dataset (so you may have to run MCMC several times).
- You then see how you did, e.g., what proportion of the 95% credible sets included the true value?

OLS versus Bayesian LASSO

			MSE		Coverage	
n	p_0	p_1	OLS	BLR	OLS	BLR
40	20	0	5.40	0.03	94.7	100.0
	15	5	5.71	3.45	93.8	96.0
	0	20	5.40	9.47	93.7	91.6
100	20	0	1.17	0.02	95.8	100.0
	15	5	1.27	0.98	94.5	95.5
	0	20	1.22	1.26	96.0	95.6

- n is the sample size.
- p_0 is the number of null covariates with $\beta_j = 0$.
- p_1 is the number of non-null covariates with $\beta_j \neq 0$.

Methods for studying frequentist properties

Conclusions:

- When the model is sparse (p_1 is small), BLR is has much smaller MSE than OLS.
- When the model is dense (p_0 is small), OLS has smaller MSE, but for large n the methods are similar.
- Both methods generally have reasonable coverage.
- BLR's coverage is low when n is small and the model is dense, i.e., when its assumptions are grossly violated.

Thank you!

Lecture 29

Bayesian Methods for Big Data

Arnab Hazra

Big data

- We are living in the era of information coming from numerous sources.
- The older statistical methods are not anymore easily implementable.
- This motivated a large number of researchers to build new methods.
- These include a large number of Bayesian methods as well.
- We will discuss some of those approaches here.

Sources of big data

- Data generated from social media
- Data collected in biomedical and healthcare informatics research such as
  - DNA sequences
  - Electronic health records
- Geospatial data generated by
  - Remote sensing
  - Laser scanning
  - Mobile mapping
  - Geo-located sensors
  - Geotagged web contents
  - Volunteered geographic information (VGI)
  - Global navigation satellite system (GNSS) tracking
- and so on...

Different directions of big data research

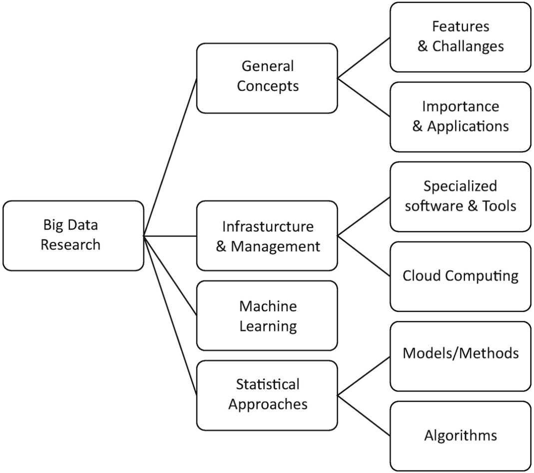


Figure: Source: Jahan et al. (2020)

Statistical Approaches to Big Data

- The importance of modeling and theoretical results are noted in the literature.
- Blind trust in algorithms without proper theoretical considerations will not result in valid outputs.
- Statistical methods play an important role for interpretability, uncertainty quantification, and reducing selection bias.
- A few important techniques pointed out in the literature are
  - Confirmatory and exploratory data analysis tools
  - Data mining methods including supervised and unsupervised learning
  - Data visualization techniques

Methodological developments

The recent methodological developments for Big Data can be divided into three broad groups

- *Subsampling*, which calculates a statistic for many subsamples taken from the data and then combine the results
- *Divide and conquer*, which breaks a dataset into smaller subsets to analyze these in parallel and then combine the results
- *Online updating of streaming data*, which performs online recursive analytical processing

Four V's of big data

- Volume
- Variety
- Veracity (biases and noise)
- Velocity

Challenges in Bayesian Approaches for Big Data

Two major challenges are

- Model construction-related
  - Models can become complex
  - A likelihood may not be available
  - Hierarchical instability
  - Parameter explosion
  - Non-identifiability of the model parameters
- Algorithm development-related
  - Scalability issues
  - Computational cost

Bayesian computational methods for Big Data

- Traditional MCMC do not scale well because they need to iterate through the full data set at each iteration to evaluate the likelihood.
- A widely used strategy here is the divide-and-conquer sampling by distributing the computation across machines.
- Here we break a massive dataset into a number of subsets, obtain posterior samples based on each subset in parallel, and finally combine the subset posterior inferences to obtain the full-posterior estimates.
- The core challenge is the recombination of sub-posterior samples to obtain true posterior samples.

Embarrassingly Parallel MCMC

- Suppose, we have N IID observations $\boldsymbol{x}_N = \{x_1, \ldots, x_N\}$.
- For the parameter vector $\boldsymbol{\theta} \in \mathbb{R}^d$, $p(\boldsymbol{\theta}|\boldsymbol{x}_N) \propto p(\boldsymbol{\theta})p(\boldsymbol{x}_N|\boldsymbol{\theta}) = p(\boldsymbol{\theta}) \prod_{i=1}^N p(x_i|\boldsymbol{\theta})$.
- We partition $\boldsymbol{x}_N$ into M subsets $\boldsymbol{x}^{n_1}, \ldots, \boldsymbol{x}^{n_M}$.
- For $m = 1, \ldots, M$ (in parallel), we sample from the sub-posterior p_m, where

$$p_m(\boldsymbol{\theta}) \propto p(\boldsymbol{\theta})^{1/M} p(\boldsymbol{x}^{n_m}|\boldsymbol{\theta}).$$

- Combine the sub-posterior samples to produce samples from an estimate of the sub-posterior density product $p_1 \times \ldots \times p_M$, which is proportional to the full-data posterior, i.e. $p_1 \times \ldots \times p_M \propto p(\boldsymbol{\theta}|\boldsymbol{x}_N)$.

Embarrassingly Parallel MCMC: Example

- Suppose, we have $X_1, \ldots, X_n | \mu \sim \text{Normal}(\mu, \sigma^2)$ IID. Here σ^2 is known.
- We choose the conjugate prior $\mu \sim \text{Normal}(\mu_0, \sigma_0^2)$.
- For $m = 1, \ldots, M$ (in parallel), we sample from the sub-posterior p_m, where

$$p_m(\mu) \propto p(\mu)^{1/M} p(\boldsymbol{x}^{n_m} | \mu).$$

- The posterior distribution for the m-th partition ($n = \sum_{i=1}^{M} n_m$) is

$$\mu | \boldsymbol{x}^{n_m} \sim \text{Normal}\left(\mu_P^{(m)} = \frac{\sigma^{-2} \text{sum}(\boldsymbol{x}^{n_m}) + \mu_0 M^{-1} \sigma_0^{-2}}{n_M \sigma^{-2} + M^{-1} \sigma_0^{-2}}, \sigma_P^{2(m)} = \frac{1}{n_M \sigma^{-2} + M^{-1} \sigma_0^{-2}} \right).$$

- Thus, $p(\mu | \boldsymbol{x}_N) \propto p_1 \times \ldots \times p_M$.
- Finally, we have

$$\mu | \boldsymbol{x}_N \sim \text{Normal}\left([\sum_{m=1}^{M} \sigma_P^{-2(m)}]^{-1} [\sum_{m=1}^{M} \sigma_P^{-2(m)} \mu_P^{(m)}], [\sum_{m=1}^{M} \sigma_P^{-2(m)}]^{-1} \right).$$

Embarrassingly Parallel MCMC: general case

- Under suitable regularity conditions, the posterior $p(\boldsymbol{\theta}|\boldsymbol{x}_N)$ is approximately equal to $\text{Normal}_d(\boldsymbol{\theta}_0, \boldsymbol{F}_N^{-1})$ (where $\boldsymbol{F}_N$ is the Fisher information matrix).
- We thus estimate each sub-posterior with $p_m(\boldsymbol{\theta}) = \text{Normal}_d(\boldsymbol{\theta}|\boldsymbol{\mu}_m, \boldsymbol{\Sigma}_m)$, where $\boldsymbol{\mu}_m$ and $\boldsymbol{\Sigma}_m$ are the sample mean and covariance, respectively, of $\boldsymbol{x}^{n_m}$.
- The product of the M sub-posterior densities will be proportional to a Gaussian pdf and our estimate of the density product function $p_1 \times \ldots \times p_M$ is

$$
\begin{aligned}
\hat{p}(\boldsymbol{\theta}|\boldsymbol{x}_N) &\propto \hat{p}_1 \times \ldots \times \hat{p}_M \\
&= \text{Normal}_d\left(\left[\sum_{m=1}^{M} \boldsymbol{\Sigma}^{(m)-1}\right]^{-1}\left[\sum_{m=1}^{M} \boldsymbol{\Sigma}^{(m)-1}\boldsymbol{\mu}^{(m)}\right], \left[\sum_{m=1}^{M} \boldsymbol{\Sigma}^{(m)-1}\right]^{-1}\right).
\end{aligned}
$$

Robust and scalable Bayes using median posterior

- Here we apply the divide-and-conquer strategy and then combine the inferences using the median of the subset posteriors.
- The median posterior (M-posterior) is constructed from the subset posteriors using Weiszfeld's algorithm, which provides a scalable algorithm for robust estimation.
- Weiszfeld's algorithm is one of the well-known and computationally efficient ways to find the geometric median in Hilbert spaces.
- For a given set of m points $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$ with each $\boldsymbol{x}_i \in \mathbb{R}^d$, the geometric median is defined as $\arg\min_{\boldsymbol{y} \in \mathbb{R}^d} \sum_{i=1}^{n} \|\boldsymbol{y} - \boldsymbol{x}_i\|_2$.
- For more details, please refer to *Robust and Scalable Bayes via a Median of Subset Posterior Measures* by Minsker et al. (2017), JMLR.

An adaptive subsampling approach

MH$\big(p(x|\theta),\, p(\theta),\, q(\theta'|\theta),\, \theta_0,\, N_{\text{iter}},\, \mathcal{X}\big)$

for $k \leftarrow 1$ **to** N_{iter}

  $\theta \leftarrow \theta_{k-1}$

  $\theta' \sim q(.|\theta)$, $u \sim \mathcal{U}_{(0,1)}$,

  $\psi(u, \theta, \theta') \leftarrow \frac{1}{n} \log \left[u \frac{p(\theta) q(\theta'|\theta)}{p(\theta') q(\theta|\theta')} \right]$

  $\Lambda_n(\theta, \theta') \leftarrow \frac{1}{n} \sum_{i=1}^{n} \log \left[\frac{p(x_i|\theta')}{p(x_i|\theta)} \right]$

  if $\Lambda_n(\theta, \theta') > \psi(u, \theta, \theta')$

    $\theta_k \leftarrow \theta'$ ▷ *Accept*

  else $\theta_k \leftarrow \theta$ ▷ *Reject*

return $(\theta_k)_{k=1,\ldots,N_{\text{iter}}}$

Figure: Source: Bardenet et al. (2014)

An algorithm for massive online streaming data

- We consider the case of a simple linear regression model $y = \boldsymbol{x}'\boldsymbol{\beta} + \varepsilon$, where $\varepsilon \sim \text{Normal}(0, \sigma^2)$.
- Suppose we choose the standard conjugate priors $\boldsymbol{\beta} \sim \text{Normal}_P(\mathbf{0}, \boldsymbol{I}_P)$ and $\sigma^2 \sim \text{Inverse-Gamma}(a, b)$.
- Let $\boldsymbol{D}^{(t)}$ be denoting the data obtained until time t.
- The full conditional posterior distributions are

$$\sigma^2|\boldsymbol{\beta}, \boldsymbol{D}^{(t)} \sim \text{Inverse-Gamma}(a_t, b_t), \quad \boldsymbol{\beta}|\sigma^2, \boldsymbol{D}^{(t)} \sim \text{Normal}_P(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t),$$

  where a_t, b_t, $\boldsymbol{\mu}_t$, and $\boldsymbol{\Sigma}_t$ follow a recursive pattern (check) in terms of sufficient statistics.
- Thus, we do not need to calculate the whole data likelihood repeatedly.
- For more details, refer to *Bayesian Conditional Density Filtering* by Qamar et al. (2015), arXiv.

Different directions of Bayesian modeling research

- Variable selection in Big Data
- Bayesian nonparametrics using various stochastic processes like
  - Gaussian processes
  - Dirichlet processes
  - Indian buffet process
  - Infinite hidden Markov models
- Bayesian Networks
- Bayesian Empirical Likelihood

Thank you!